Financial Econometrics Practice TS #2 (Forecasting)

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1 Load Packages and Data

```r
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.5.2
```

```r
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.5.2

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(lmtest)
```

```
## Loading required package: zoo

## Warning: package 'zoo' was built under R version 4.5.2

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(nortest)
library(MASS)
```

Read the data as a time series object. The data is the number of individuals registered as unemployed in SEPE offices since January 2000 available at https://expinterweb.mites.gob.es/series/. Select window starting 2000.

```r
series <- window(
  ts(read.table("unemploy.dat", dec = ",") / 1e6,
     start = 1996, freq = 12),
  start = c(2000),
  end = c(2025, 12)
)

# Display properties
cat("Class:", class(series), "\n")
```

```
## Class: ts
```

```r
cat("Length:", length(series), "observations\n")
```

```
## Length: 312 observations
```

```r
cat("Start:", start(series)[1], "Period", start(series)[2], "\n")
```

```
## Start: 2000 Period 1
```

```r
cat("End:", end(series)[1], "Period", end(series)[2], "\n")
```

```
## End: 2025 Period 12
```

```r
cat("Frequency:", frequency(series), "\n\n")
```

```
## Frequency: 12
```

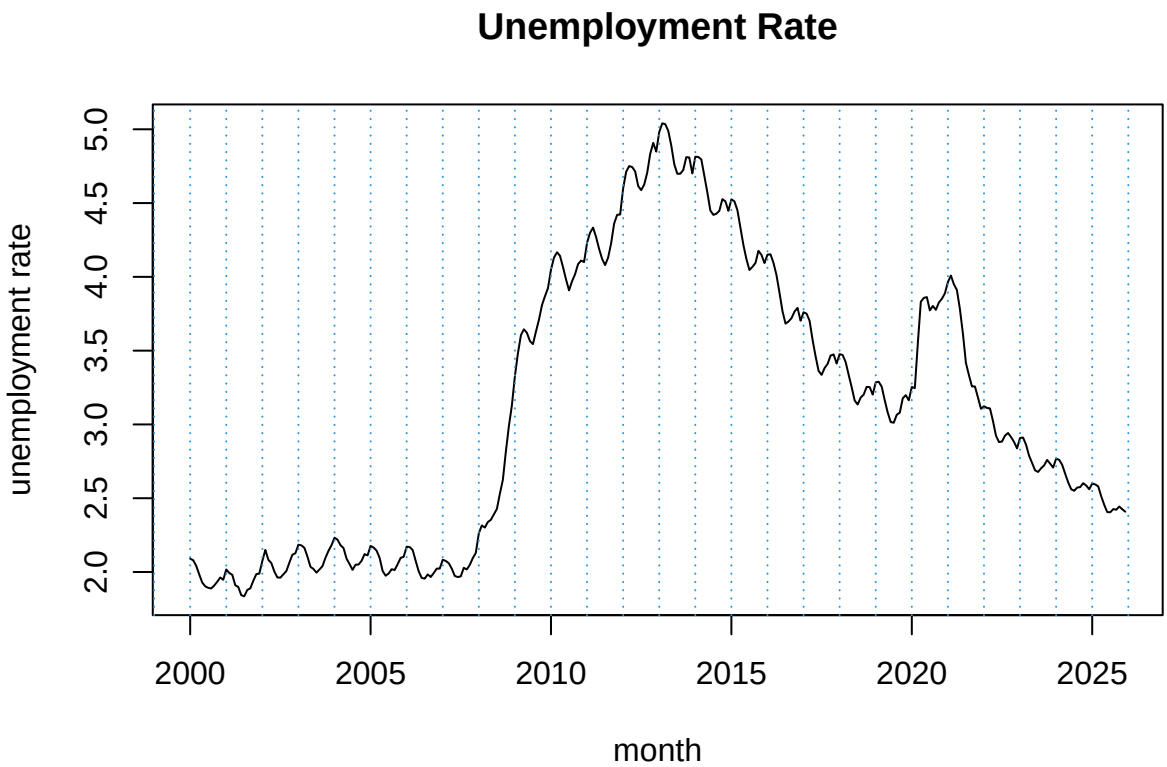
```r
# show values
round(series, 3)
```

```
##        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 2000 2.090 2.080 2.042 1.984 1.929 1.903 1.893 1.888 1.908 1.934 1.963 1.947
## 2001 2.017 1.993 1.981 1.910 1.898 1.843 1.836 1.879 1.889 1.941 1.986 1.989
## 2002 2.075 2.150 2.083 2.060 2.003 1.963 1.962 1.984 2.007 2.065 2.117 2.127
## 2003 2.185 2.180 2.163 2.104 2.036 2.020 1.996 2.017 2.040 2.097 2.143 2.181
## 2004 2.233 2.219 2.182 2.162 2.091 2.054 2.014 2.050 2.051 2.076 2.121 2.113
## 2005 2.177 2.165 2.145 2.096 2.007 1.975 1.989 2.019 2.013 2.053 2.096 2.103
## 2006 2.172 2.169 2.149 2.076 2.005 1.960 1.955 1.984 1.966 1.993 2.023 2.023
## 2007 2.083 2.075 2.059 2.023 1.973 1.966 1.970 2.028 2.017 2.049 2.094 2.130
## 2008 2.262 2.315 2.301 2.339 2.354 2.390 2.427 2.530 2.625 2.818 2.989 3.129
## 2009 3.328 3.482 3.605 3.645 3.620 3.565 3.544 3.629 3.709 3.808 3.869 3.924
## 2010 4.048 4.131 4.167 4.142 4.066 3.982 3.909 3.970 4.018 4.086 4.110 4.100
## 2011 4.231 4.299 4.334 4.269 4.190 4.122 4.080 4.131 4.227 4.361 4.420 4.422
## 2012 4.600 4.712 4.751 4.744 4.714 4.615 4.587 4.626 4.705 4.834 4.908 4.849
## 2013 4.981 5.040 5.035 4.989 4.891 4.764 4.699 4.699 4.724 4.811 4.809 4.701
## 2014 4.814 4.812 4.796 4.684 4.572 4.450 4.420 4.428 4.448 4.527 4.512 4.448
## 2015 4.526 4.512 4.452 4.333 4.215 4.120 4.046 4.068 4.094 4.176 4.149 4.094
## 2016 4.151 4.153 4.095 4.011 3.891 3.767 3.683 3.697 3.720 3.765 3.790 3.703
## 2017 3.760 3.751 3.702 3.573 3.461 3.363 3.336 3.382 3.410 3.467 3.474 3.413
## 2018 3.477 3.470 3.423 3.336 3.252 3.162 3.135 3.182 3.203 3.255 3.253 3.202
## 2019 3.286 3.289 3.255 3.164 3.079 3.016 3.011 3.066 3.080 3.178 3.198 3.164
## 2020 3.254 3.246 3.548 3.831 3.858 3.863 3.773 3.803 3.776 3.826 3.851 3.888
## 2021 3.964 4.009 3.950 3.911 3.781 3.614 3.416 3.334 3.258 3.257 3.183 3.106
## 2022 3.123 3.112 3.109 3.023 2.923 2.881 2.884 2.924 2.942 2.915 2.881 2.838
## 2023 2.908 2.911 2.862 2.788 2.739 2.689 2.678 2.703 2.722 2.759 2.735 2.707
## 2024 2.768 2.760 2.727 2.667 2.608 2.561 2.550 2.572 2.575 2.602 2.586 2.561
## 2025 2.599 2.593 2.580 2.513 2.455 2.406 2.405 2.427 2.422 2.444 2.425 2.409
```

```r
# constants
series_name <- "Unemployment Rate"
start_year <- 2000
start_period <- 1
freq <- 12
period_length <- 12
xlabel <- "month"
ylabel <- "unemployment rate"
```

some obvious yearly seasonality, but SARIMA will handle that

```r
plot(series, main=series_name, xlab=xlabel, ylab=ylabel)
abline(v=1996:2026,col=4,lty=3)
```



To model, we first achieve variance constancy and let SARIMA handle the seasonality and mean stability.

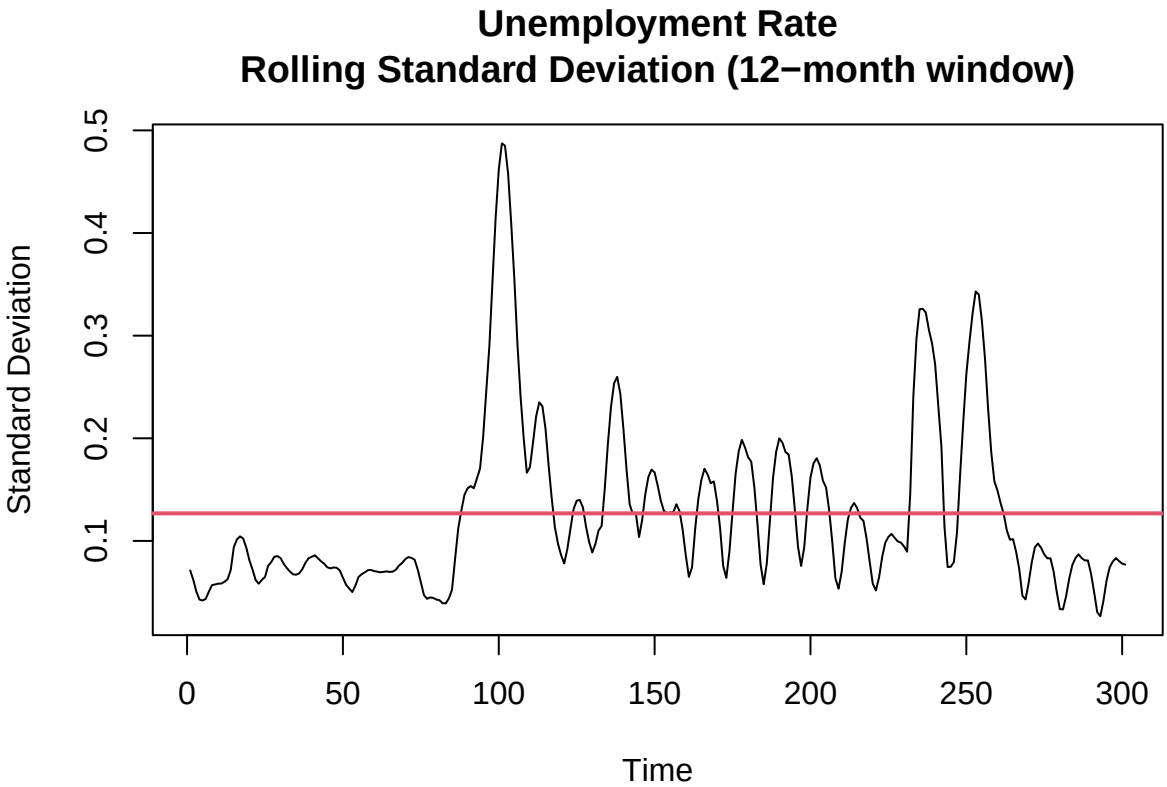
2 Transformation to Variance Constancy

use rolling window stdev better than boxplot, with heteroskedasticity

```
#function to plot rolling standard deviation
plot_rolling_std <- function(series, window_size, name="Series") {
  rolling_std <- sapply(1:(length(series) - window_size + 1), function(i) {
    sd(series[i:(i + window_size - 1)])
  })

  plot(rolling_std, type = "l",
    main = paste(name, "\nRolling Standard Deviation (",
           window_size, "-month window)", sep = ""),
    xlab = "Time", ylab = "Standard Deviation")
  abline(h=mean(rolling_std), col=2, lwd=2)
}
```

2.1 Rolling standard deviation

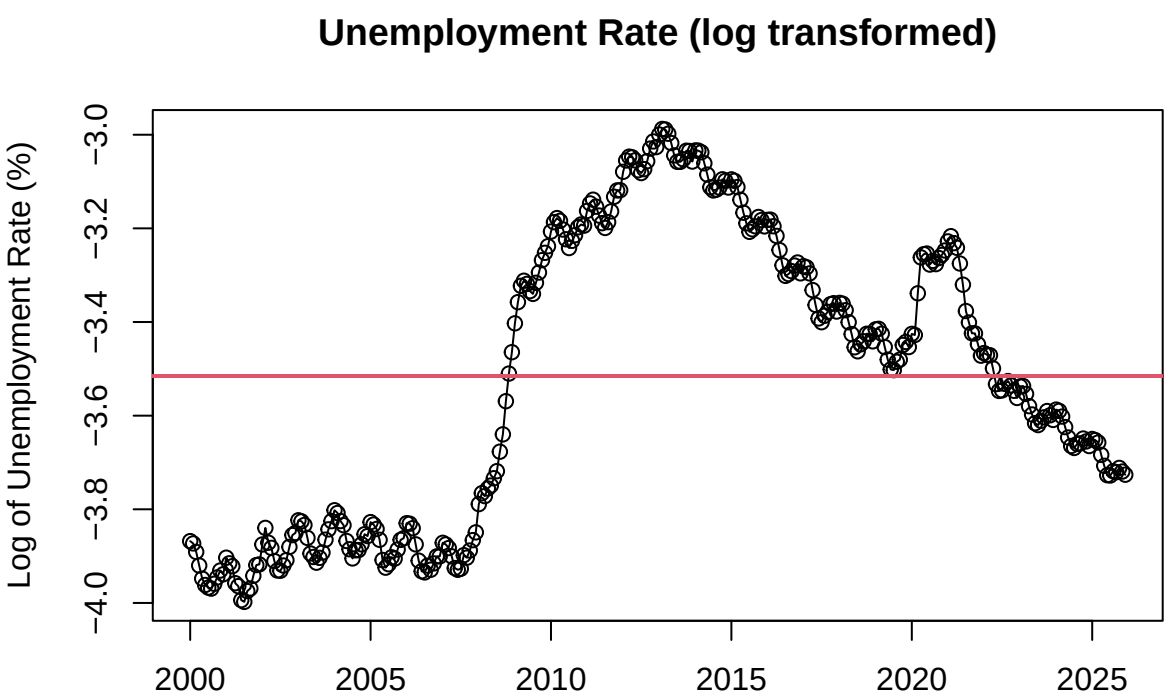
```
plot_rolling_std(series, 12, series_name)
```



##

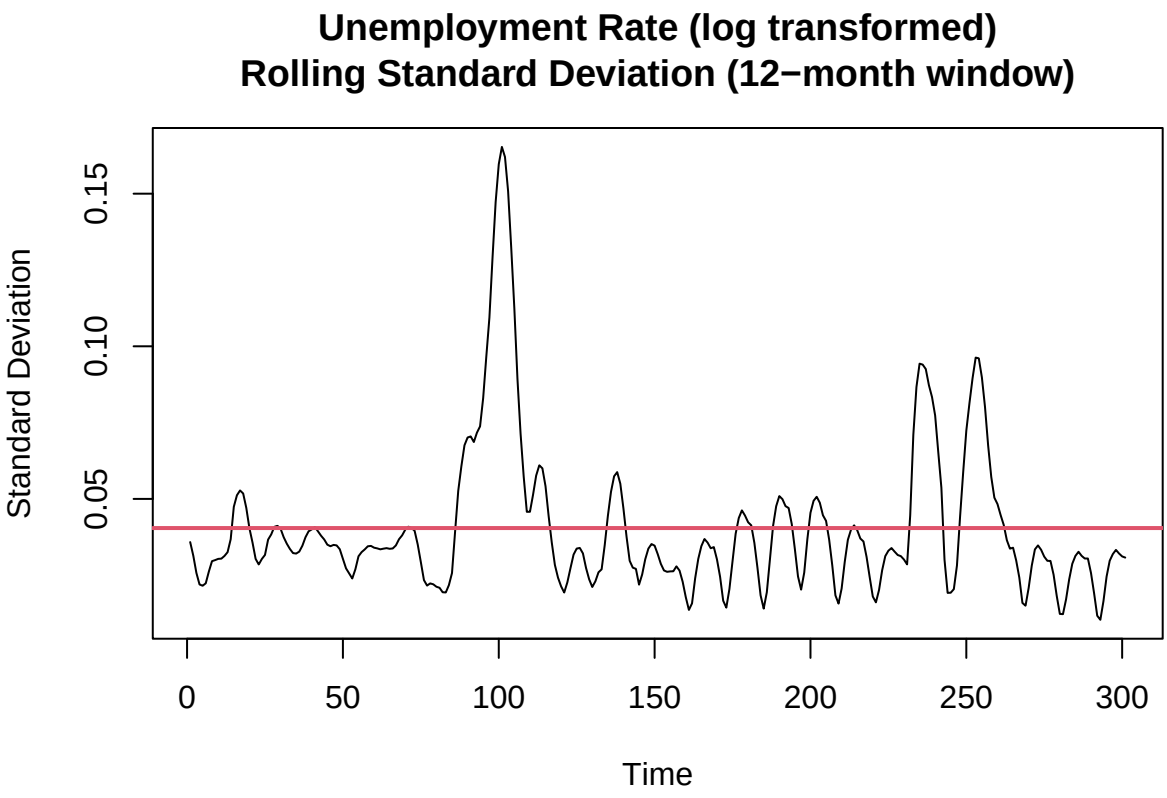
LN Transformation seemed to handle mean stability and seasonality

```
ln_transformed <- log(series / 100)
plot(ln_transformed, type="o",
  main=paste(series_name, "(log transformed)"),
  ylab=paste("Log of Unemployment Rate (%)", sep=""),
  xlab="")
abline(h=mean(ln_transformed), col=2, lwd=2)
```



still with obvious heteroskedasticity here

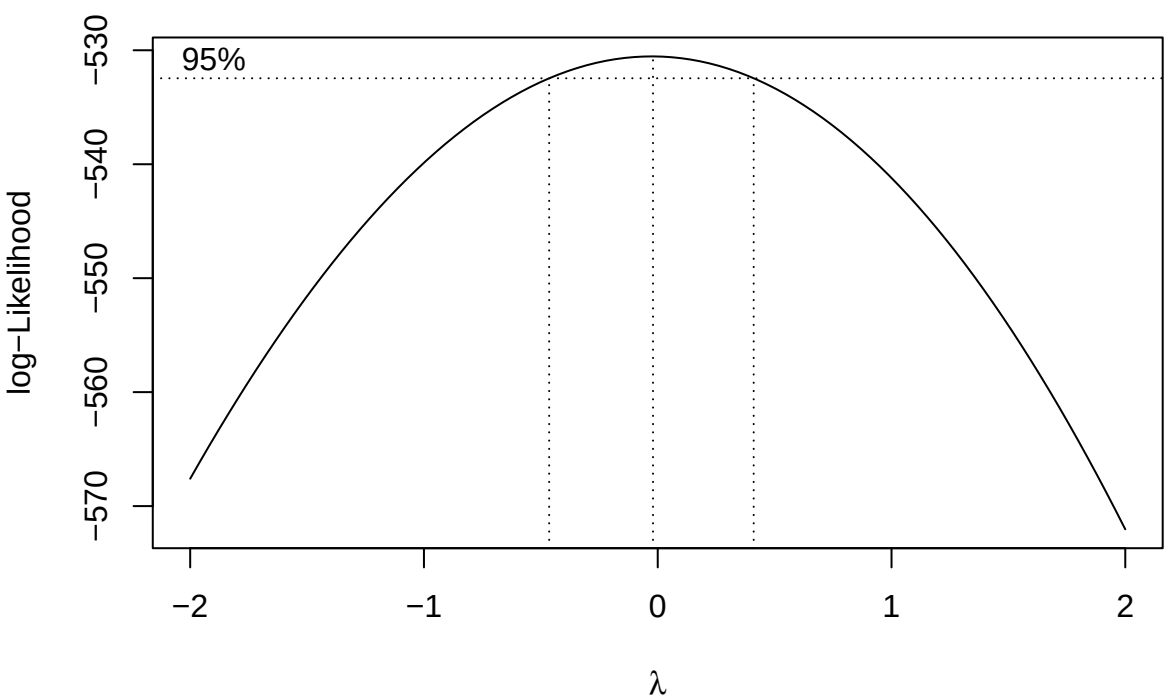
```
plot_rolling_std(ln_transformed, 12, paste(series_name, "(log transformed)"))
```



##

Box Cox Transformation

```
# Box-Cox transformation to find optimal lambda for variance stabilization
boxcox_result <- boxcox(series ~ 1, lambda = seq(-2, 2, by = 0.1))
```

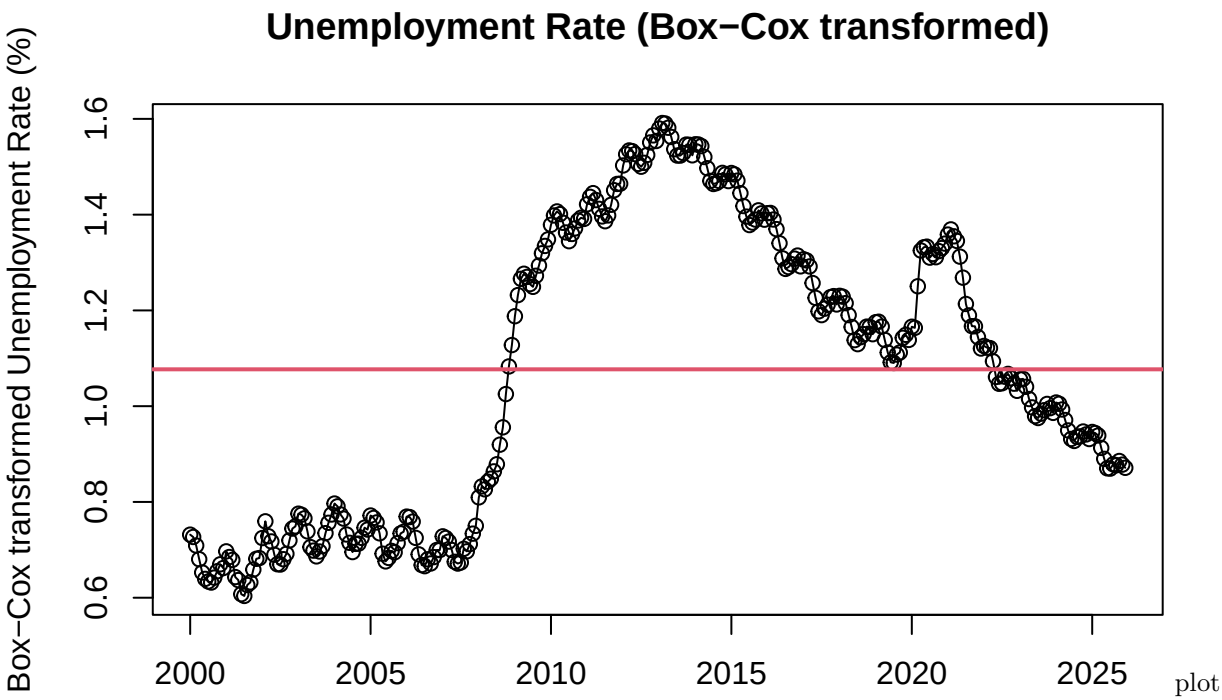


```r
optimal_lambda <- boxcox_result$x[which.max(boxcox_result$y)]
cat("Optimal lambda for Box-Cox transformation:", optimal_lambda, "\n")
```

```
## Optimal lambda for Box-Cox transformation: -0.02020202
```

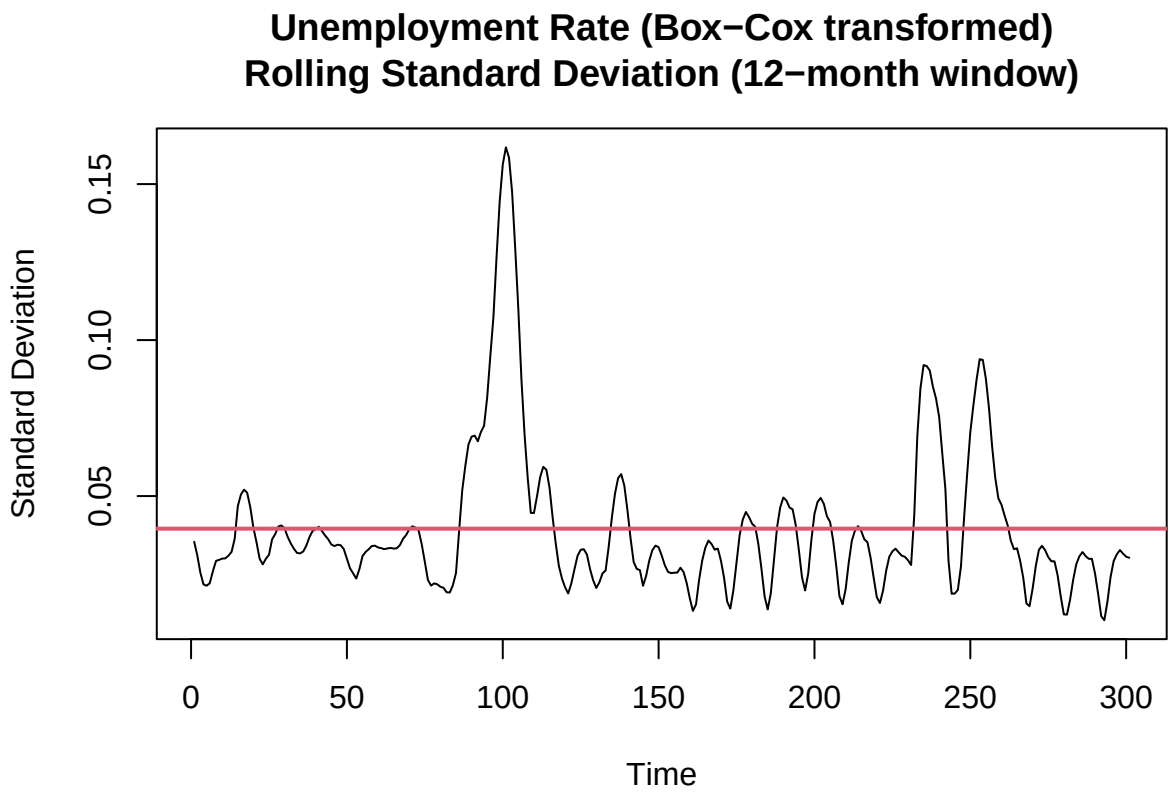
```r
# Apply Box-Cox transformation with optimal lambda
bc_transformed <- (series^optimal_lambda - 1) / optimal_lambda
```

```r
plot(bc_transformed, type="o",
  main=paste(series_name, "(Box-Cox transformed)"),
  ylab=paste("Box-Cox transformed Unemployment Rate (%)", sep=""),
  xlab="")
abline(h=mean(bc_transformed), col=2, lwd=2)
```



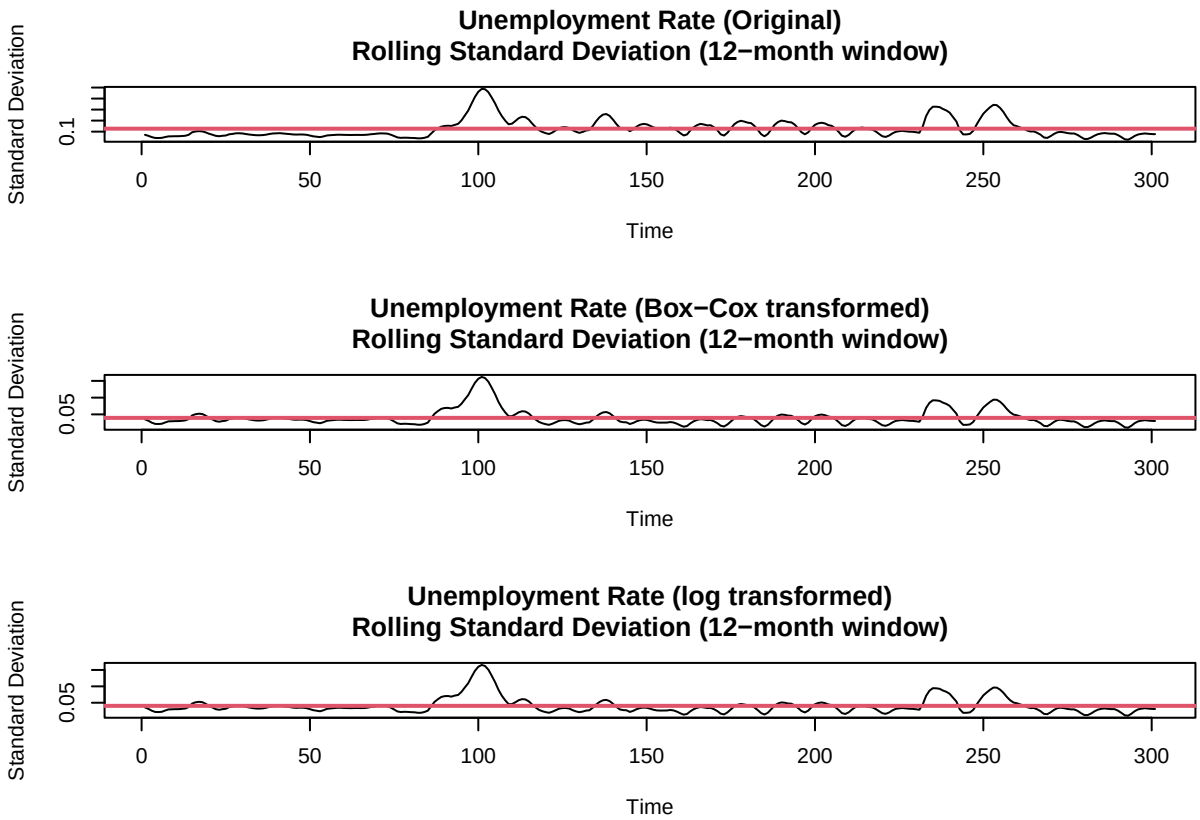
plot

```r
plot_rolling_std(bc_transformed, 12, paste(series_name, "(Box-Cox transformed)"))
```



plot all three transformations together for comparison

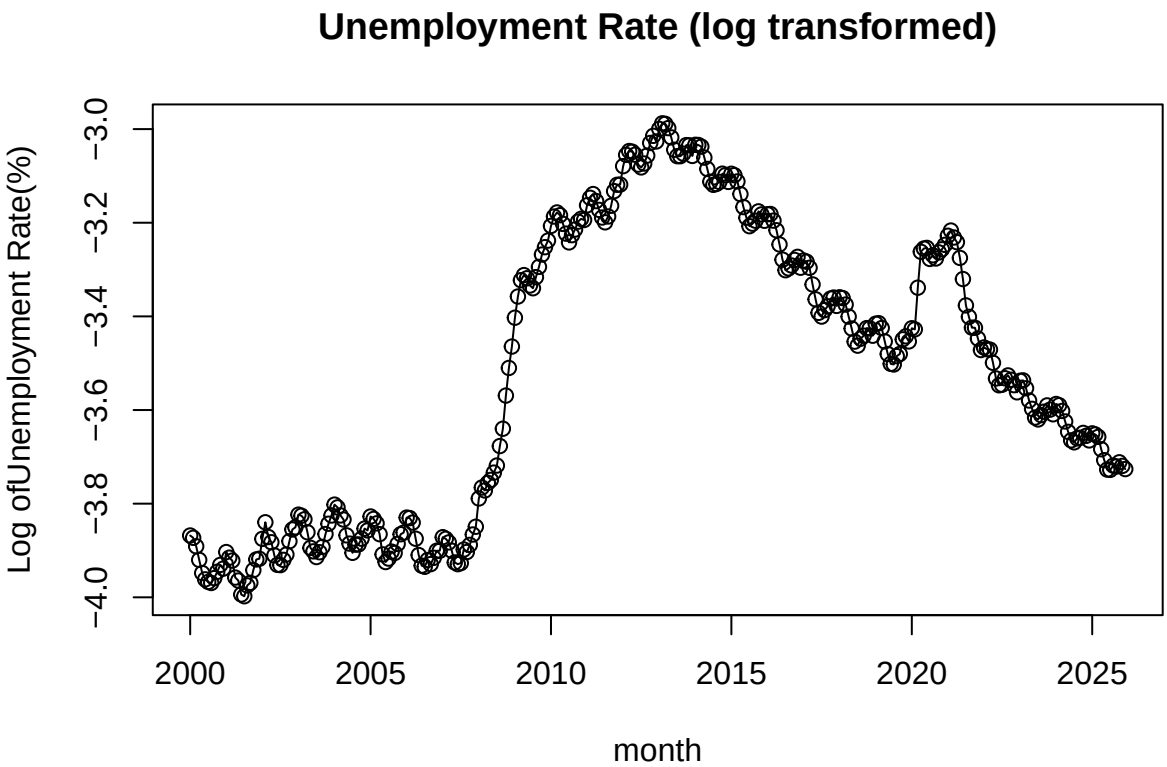
```
par(mfrow=c(3,1))
plot_rolling_std(series, 12, paste(series_name, "(Original)"))
plot_rolling_std(bc_transformed, 12, paste(series_name, "(Box-Cox transformed)"))
plot_rolling_std(ln_transformed, 12, paste(series_name, "(log transformed)"))
```



#

Prep for SARIMA

```
par(mfrow=c(1,1))
working_series <- ln_transformed
working_series_name <- paste(series_name, "(log transformed)")
ylabel <- paste("Log of", series_name, "(%)", sep="")
plot(working_series, type="o", main=working_series_name, ylab=ylabel, xlab=xlabel)
```



ACF

PACF to see potential sarima models

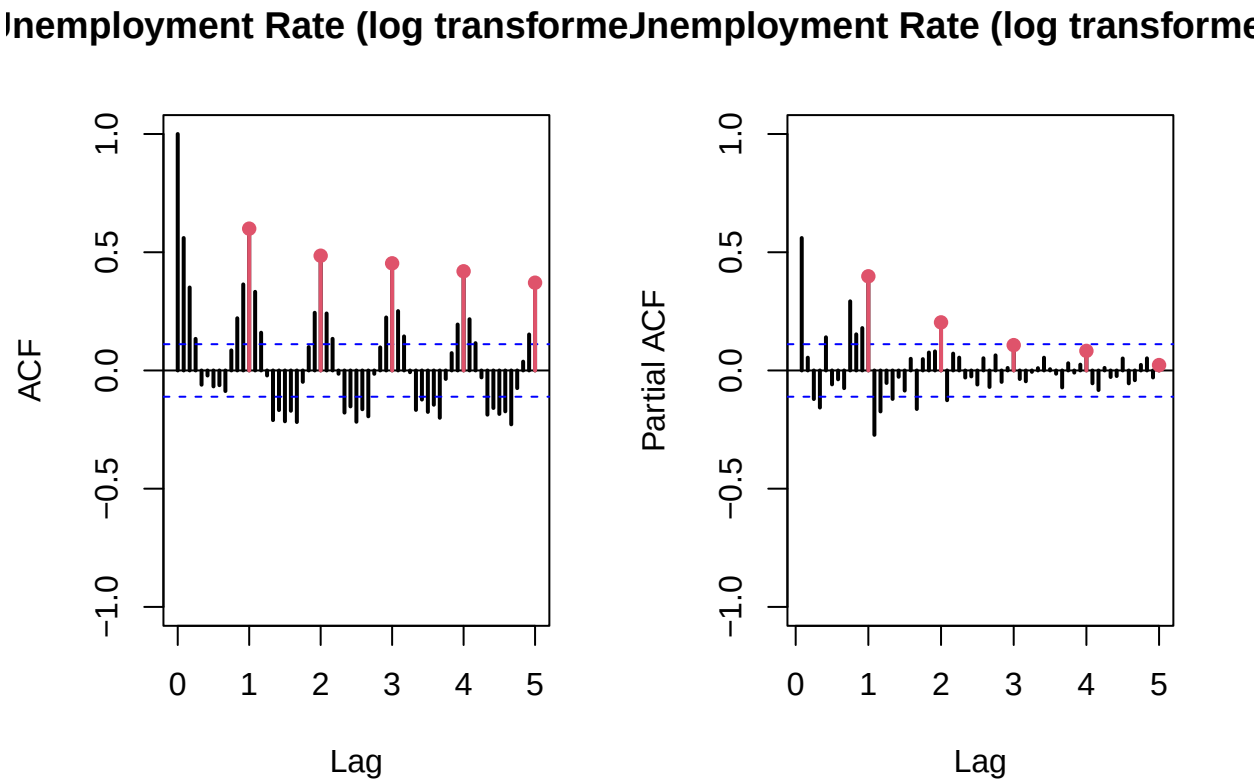
definitely need differencing

```
par(mfrow=c(1,2))
acf(working_series, ylim=c(-1,1), lag.max=max(60, 3*freq),
    lwd=2, main=paste("ACF:", working_series_name),
)
pacf(working_series, ylim=c(-1,1), lag.max=max(60, 3*freq),
     lwd=2, main=paste("PACF:", working_series_name),
     col=c(2, rep(1, 59)))
```

```
par(mfrow=c(1,1))
```

Do first differencing to determine seasonality and ARMA structure

```
diff_series <- diff(working_series)
par(mfrow=c(1,2))
lag_limit <- max(60, 3*freq)
seasonal_lags <- seq(freq, lag_limit, by=freq)

acf_obj <- acf(diff_series, ylim=c(-1,1), lag.max=lag_limit,
         lwd=2, col=1, main=paste("ACF:", working_series_name, "(1st Diff)"))
acf_lags <- as.numeric(acf_obj$lag)[-1]
acf_vals <- as.numeric(acf_obj$acf)[-1]
acf_idx <- which(round(acf_lags * freq) %in% seasonal_lags)
segments(acf_lags[acf_idx], 0, acf_lags[acf_idx], acf_vals[acf_idx], col=2, lwd=2)
points(acf_lags[acf_idx], acf_vals[acf_idx], col=2, pch=16)

pacf_obj <- pacf(diff_series, ylim=c(-1,1), lag.max=lag_limit,
         lwd=2, col=1, main=paste("PACF:", working_series_name, "(1st Diff)"))
pacf_lags <- as.numeric(pacf_obj$lag)
pacf_vals <- as.numeric(pacf_obj$acf)
pacf_idx <- which(round(pacf_lags * freq) %in% seasonal_lags)
segments(pacf_lags[pacf_idx], 0, pacf_lags[pacf_idx], pacf_vals[pacf_idx], col=2, lwd=2)
points(pacf_lags[pacf_idx], pacf_vals[pacf_idx], col=2, pch=16)
```



We should do seasonality 12 for both AR and MA, and then up to AR(5) and MA(3) for the non-seasonal part, based on the ACF and PACF patterns.

```
working_series <- diff_series
```

3 SARIMA Modeling

3.1 SARIMA function

```
analyze_sarima <- function(series, order, seasonal) {

  # --- Setup ---
  s <- frequency(series)
  n <- length(series)
  model_name <- paste0("ARIMA(", paste(order, collapse=","), ")(",
                       paste(seasonal$order, collapse=","), ")", s)

  cat("\n==================================================================\n")
  cat("  ANALYSIS FOR MODEL:", model_name, "\n")
  cat("==================================================================\n")

  # ----------------------------------------------------------------------
  # 1. Estimation & Significance Check
  # ----------------------------------------------------------------------
  fit_full <- arima(series, order = order, seasonal = seasonal)

  diag_var <- diag(fit_full$var.coef)
  if (any(diag_var <= 0)) {
    cat("\n  CRITICAL WARNING: The Hessian matrix is unstable (Negative Variances).\n")
```

```r
  se_vec <- ifelse(diag_var > 0, sqrt(diag_var), NA)
} else {
  se_vec <- sqrt(diag_var)
}

t_vec  <- fit_full$coef / se_vec
p_vec  <- 2 * (1 - pnorm(abs(t_vec)))

coeftest <- data.frame(
  Estimate = fit_full$coef,
  Std.Error = se_vec,
  t_stat = t_vec,
  p_value = p_vec
)

coeftest$Significant <- ifelse(!is.na(coeftest$p_value) &
                                 (coeftest$p_value < 0.05 | abs(coeftest$t_stat) > 2),
                               "Yes", "No")

cat("\n[1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):\n")
print(round(coeftest[,1:4], 4))
cat("\nSignificance Flags:\n")
print(coeftest[,5, drop=FALSE])

# ----------------------------------------------------------------------
# 2. Residual Analysis (Diagnostics)
# ----------------------------------------------------------------------
resid <- residuals(fit_full)

# A. Causality / Invertibility Roots (Min Modulus)
# AR Roots
ar_poly <- fit_full$model$phi
if (length(ar_poly) > 0 && any(ar_poly != 0)) {
  ar_roots_mod <- Mod(polyroot(c(1, -ar_poly)))
  min_ar_root  <- min(ar_roots_mod)
} else {
  min_ar_root  <- NA
}

# MA Roots
ma_poly <- fit_full$model$theta
if (length(ma_poly) > 0 && any(ma_poly != 0)) {
  ma_roots_mod <- Mod(polyroot(c(1, ma_poly)))
  min_ma_root  <- min(ma_roots_mod)
} else {
  min_ma_root  <- NA
}

is_causal     <- ifelse(!is.na(min_ar_root), min_ar_root > 1, TRUE)
is_invertible <- ifelse(!is.na(min_ma_root), min_ma_root > 1, TRUE)

cat("\n[2] DIAGNOSTICS:\n")
cat("Min AR Root Modulus: ", min_ar_root, "\n")
```

```r
cat("Min MA Root Modulus: ", min_ma_root, "\n")

# B. Plots
par(mfrow=c(2,2), mar=c(4,4,3,2))
plot(resid, main="Residuals", ylab="Error")
abline(h=0, col="gray"); abline(h=c(-3*sd(resid), 3*sd(resid)), lty=2, col="red")
scatter.smooth(sqrt(abs(resid)), main="Variance Check", ylab="sqrt(|Resid|)", lpars=list(col="red"))
qqnorm(resid); qqline(resid, col="red")
hist(resid, breaks=20, freq=FALSE, main="Histogram"); curve(dnorm(x, mean=mean(resid), sd=sd(resid)),

# C. Tests
sw_test <- shapiro.test(resid)
lb_test <- Box.test(resid, lag = 20, type = "Ljung-Box")

# ---------------------------------------------------------------------
# 3. Forecasting & Cross-Validation
# ---------------------------------------------------------------------
train_end <- c(floor(time(series)[n-12]), cycle(series)[n-12])
train_set <- window(series, end = train_end)
test_set  <- window(series, start = time(series)[n-11])

fit_train <- arima(train_set, order = order, seasonal = seasonal)
fc <- predict(fit_train, n.ahead = 12)

fc_orig <- exp(fc$pred); test_orig <- exp(test_set)
lower_bound <- exp(fc$pred - 1.96 * fc$se); upper_bound <- exp(fc$pred + 1.96 * fc$se)

# ---------------------------------------------------------------------
# 4. Summary Table Generation
# ---------------------------------------------------------------------
summary_row <- data.frame(
  Model_Name      = model_name,
  AIC             = fit_full$aic,
  BIC             = BIC(fit_full),
  Sigma2          = fit_full$sigma2,

  Resid_Norm_PVal = sw_test$p.value,
  Normal       = ifelse(sw_test$p.value > 0.05, "Yes", "No"),

  Resid_Ind_PVal  = lb_test$p.value,
  Independent  = ifelse(lb_test$p.value > 0.05, "Yes", "No"),

  # Positioned as requested: Root value to the left of the Boolean status
  Min_AR_Root     = min_ar_root,
  Causal          = ifelse(is_causal, "Yes", "No"),

  Min_MA_Root     = min_ma_root,
  Invertible      = ifelse(is_invertible, "Yes", "No"),

  RMSE            = sqrt(mean((test_orig - fc_orig)^2)),
  RMSPE           = sqrt(mean(((test_orig - fc_orig) / test_orig)^2)) * 100,
  MAE             = mean(abs(test_orig - fc_orig)),
  MAPE_Pct        = mean(abs((test_orig - fc_orig) / test_orig)) * 100,
```

```r
    Mean_CI_Width   = mean(upper_bound - lower_bound)
  )

  cat("\n[3] PERFORMANCE SUMMARY:\n")
  print(t(summary_row))

  return(summary_row)
}
```

3.2 Comparison Function

```r
# -- COMPARATIVE ANALYSIS FUNCTION --


compare_sarima_models <- function(series, models_list) {

  cat("\n--- Starting Automated Model Comparison ---\n")

  # This will store the summary row for each successful model
  comparison_results <- list()

  # Loop through each model specification in the list
  for (i in seq_along(models_list)) {
    spec <- models_list[[i]]

    cat(paste0("\nProcessing Model ", i, " of ", length(models_list), "...\n"))

    # Use tryCatch to prevent the loop from stopping if a model fails to converge
    result <- tryCatch({
      analyze_sarima(series = series,
                     order = spec$order,
                     seasonal = spec$seasonal)
    }, error = function(e) {
      cat(paste0("  Error in Model ", i, ": ", e$message, "\n"))
      return(NULL) # Return NULL so we can skip it
    })

    # If the model was successful, add it to our list
    if (!is.null(result)) {
      comparison_results[[length(comparison_results) + 1]] <- result
    }
  }

  # Combine all the rows into one master table
  final_table <- do.call(rbind, comparison_results)

  # Sort by BIC by default (it penalizes complexity more than AIC)
  final_table <- final_table[order(final_table$BIC), ]

  cat("\n--- Comparison Complete ---\n")
  return(final_table)
}
```

3.3 Models

```r
# All possible non-seasonal ARMA(p,q) with p <= 5 and q <= 3
# Seasonal period fixed at 12
model_grid <- expand.grid(p = 1:5, q = 1:3)

my_models <- lapply(seq_len(nrow(model_grid)), function(i) {
  list(
    order = c(model_grid$p[i], 0, model_grid$q[i]),
    seasonal = list(order = c(1,0,1), period = 12)
  )
})

# Print all model names
model_names <- paste0(
  "ARIMA(", model_grid$p, ",0,", model_grid$q, ")",
  "(1,0,1)[12]"
)
print(model_names)
```

```
##  [1] "ARIMA(1,0,1)(1,0,1)[12]" "ARIMA(2,0,1)(1,0,1)[12]"
##  [3] "ARIMA(3,0,1)(1,0,1)[12]" "ARIMA(4,0,1)(1,0,1)[12]"
##  [5] "ARIMA(5,0,1)(1,0,1)[12]" "ARIMA(1,0,2)(1,0,1)[12]"
##  [7] "ARIMA(2,0,2)(1,0,1)[12]" "ARIMA(3,0,2)(1,0,1)[12]"
##  [9] "ARIMA(4,0,2)(1,0,1)[12]" "ARIMA(5,0,2)(1,0,1)[12]"
## [11] "ARIMA(1,0,3)(1,0,1)[12]" "ARIMA(2,0,3)(1,0,1)[12]"
## [13] "ARIMA(3,0,3)(1,0,1)[12]" "ARIMA(4,0,3)(1,0,1)[12]"
## [15] "ARIMA(5,0,3)(1,0,1)[12]"
```

show a few of the models to confirm they are correct

```r
head(my_models, 1)
```

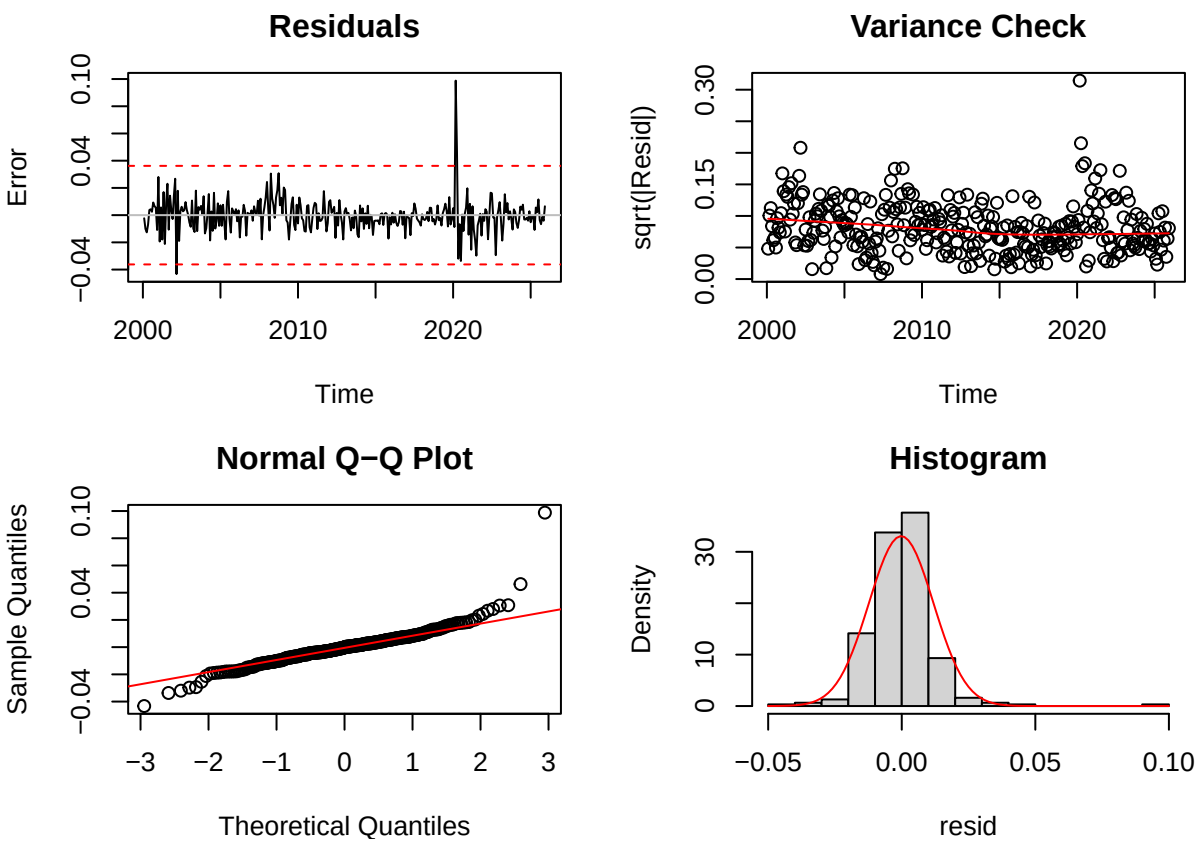
```
## [[1]]
## [[1]]$order
## [1] 1 0 1
##
## [[1]]$seasonal
## [[1]]$seasonal$order
## [1] 1 0 1
##
## [[1]]$seasonal$period
## [1] 12
```

3.4 Run all models

```r
# Run Analysis
comparison_table <- compare_sarima_models(working_series, my_models)
```

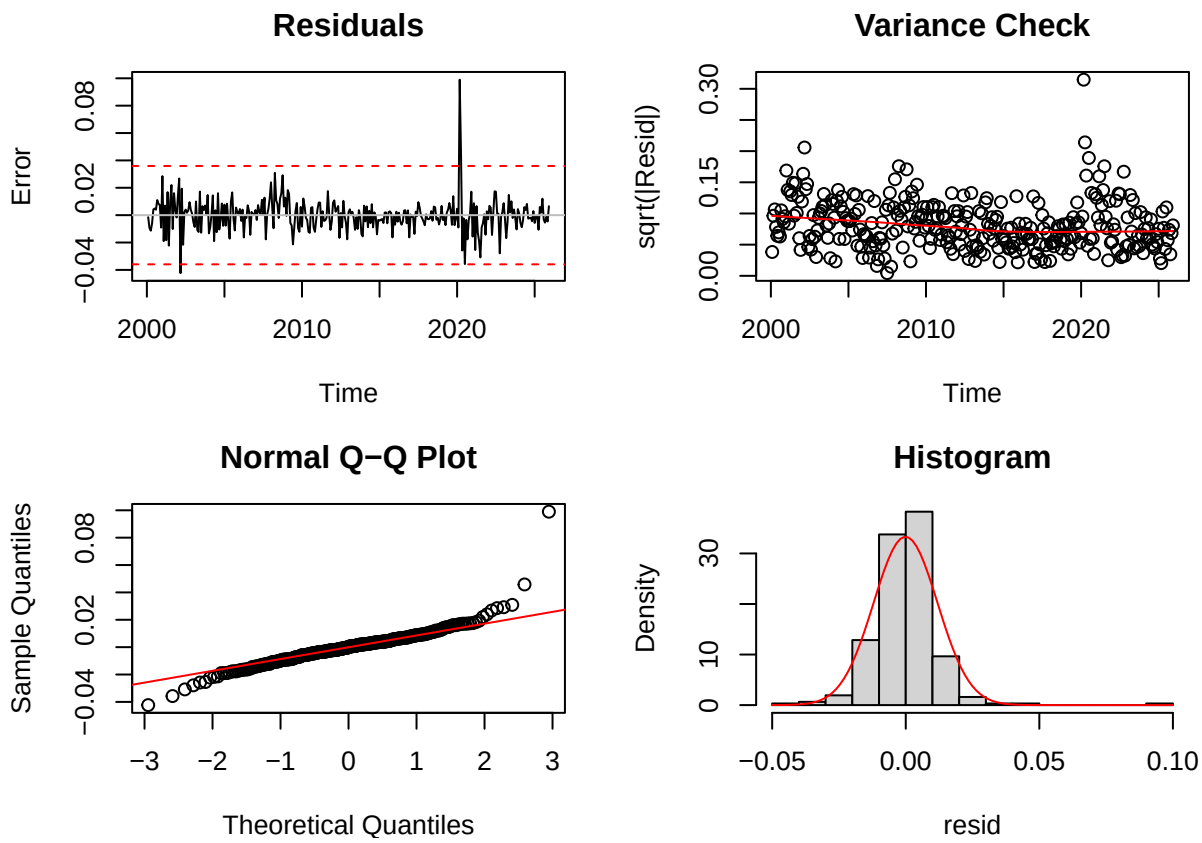
```
##
## --- Starting Automated Model Comparison ---
##
## Processing Model 1 of 15...
##
## ==================================================================
```

```
##   ANALYSIS FOR MODEL: ARIMA(1,0,1)(1,0,1)12
## ================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##            Estimate Std.Error   t_stat p_value
## ar1          0.7467    0.0633  11.7909  0.0000
## ma1         -0.2189    0.0977  -2.2402  0.0251
## sar1         0.9914    0.0065 152.6144  0.0000
## sma1        -0.8536    0.0479 -17.8350  0.0000
## intercept   -0.0006    0.0117  -0.0493  0.9607
##
## Significance Flags:
##            Significant
## ar1                Yes
## ma1                Yes
## sar1               Yes
## sma1               Yes
## intercept           No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000721
## Min MA Root Modulus:  1.013275
```



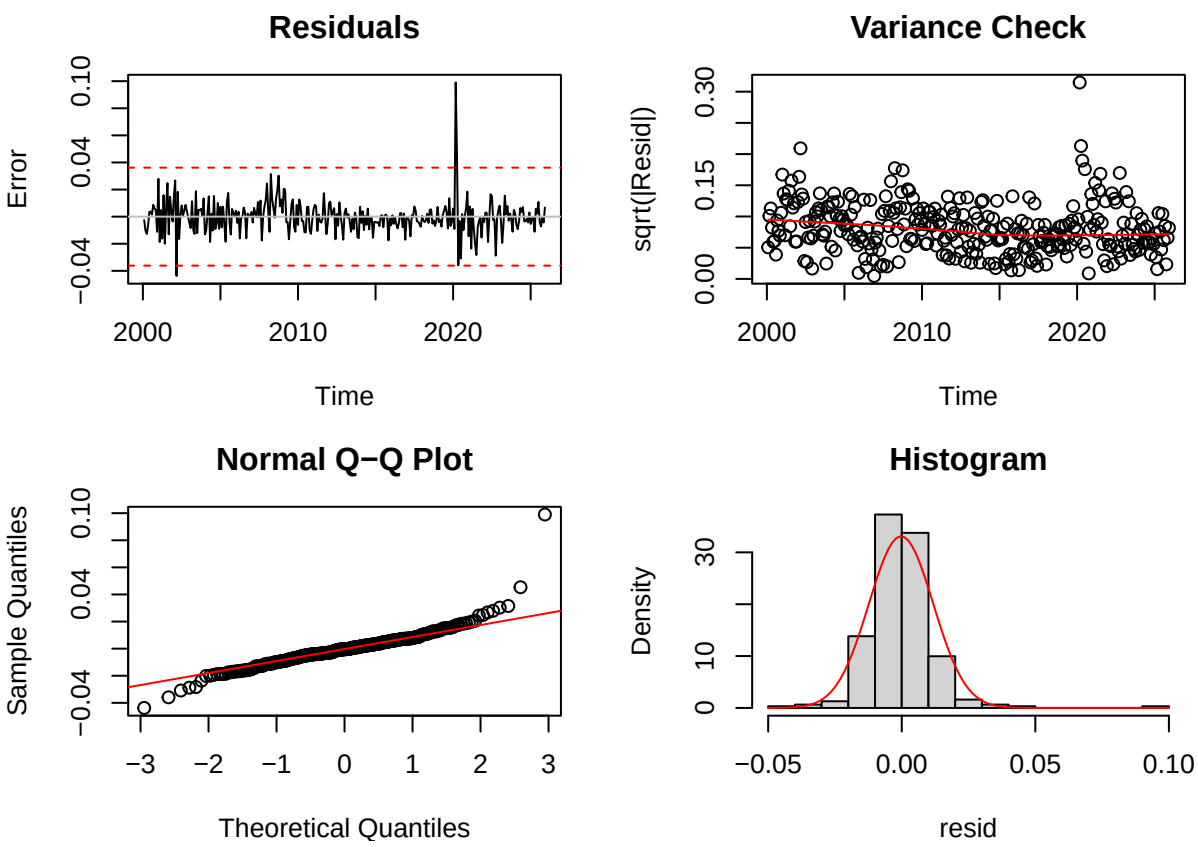

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(1,0,1)(1,0,1)12"
## AIC              "-1833.625"
```

```
## BIC             "-1811.186"
## Sigma2          "0.0001452986"
## Resid_Norm_PVal "1.170265e-14"
## Normal          "No"
## Resid_Ind_PVal  "0.3330666"
## Independent     "Yes"
## Min_AR_Root     "1.000721"
## Causal          "Yes"
## Min_MA_Root     "1.013275"
## Invertible      "Yes"
## RMSE            "0.005897894"
## RMSPE           "0.5961513"
## MAE             "0.004715277"
## MAPE_Pct        "0.4746397"
## Mean_CI_Width   "0.05873444"
##
## Processing Model 2 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(2,0,1)(1,0,1)12
## ==================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         1.2911    0.1290  10.0076  0.0000
## ar2        -0.3483    0.1005  -3.4639  0.0005
## ma1        -0.7649    0.1067  -7.1662  0.0000
## sar1        0.9930    0.0054 183.5141  0.0000
## sma1       -0.8650    0.0457 -18.9451  0.0000
## intercept  -0.0018    0.0160  -0.1158  0.9078
##
## Significance Flags:
##           Significant
## ar1               Yes
## ar2               Yes
## ma1               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000582
## Min MA Root Modulus:  1.012163
```



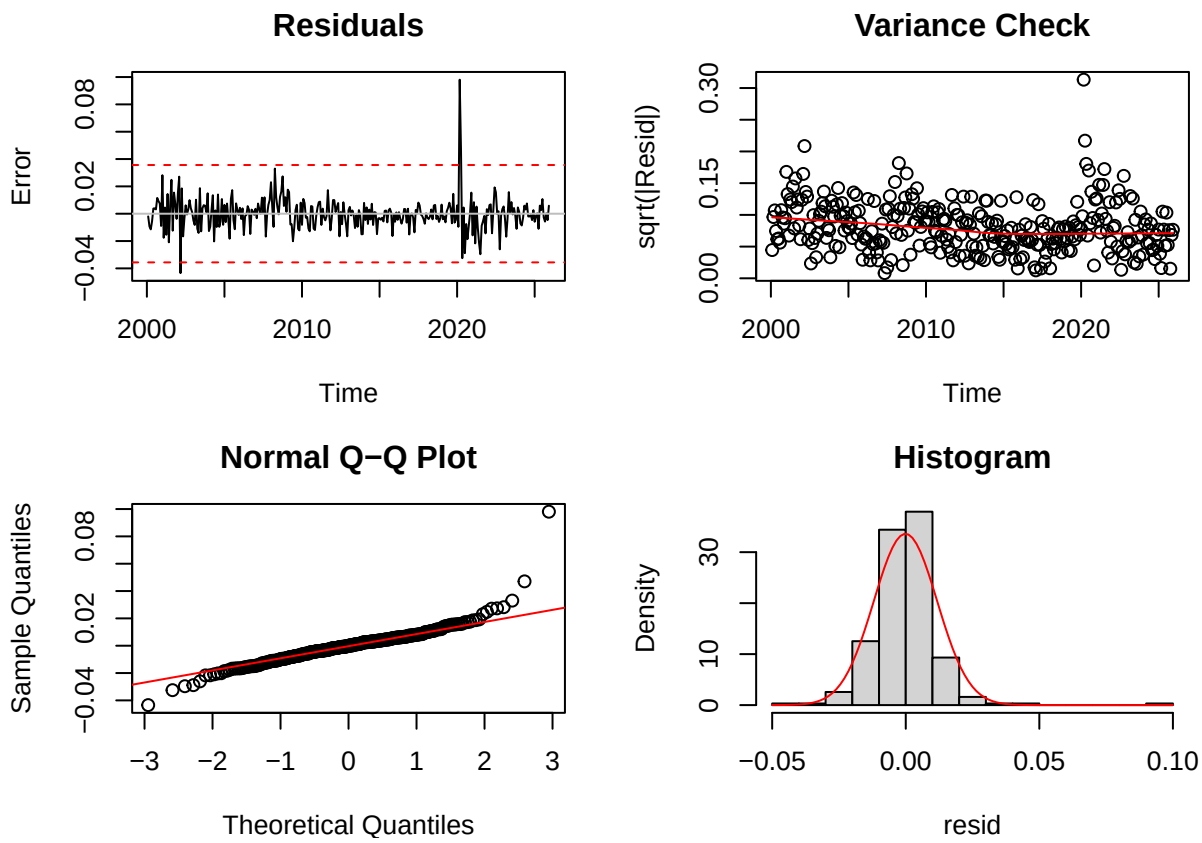

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(2,0,1)(1,0,1)12"
## AIC              "-1835.071"
## BIC              "-1808.892"
## Sigma2           "0.0001429969"
## Resid_Norm_PVal "9.511915e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.4642005"
## Independent      "Yes"
## Min_AR_Root      "1.000582"
## Causal           "Yes"
## Min_MA_Root      "1.012163"
## Invertible       "Yes"
## RMSE             "0.005911938"
## RMSPE            "0.5968776"
## MAE              "0.004688363"
## MAPE_Pct         "0.4715944"
## Mean_CI_Width    "0.05737218"
##
## Processing Model 3 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(3,0,1)(1,0,1)12
## ==================================================================
##
```

```
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1        -0.2060    0.3399  -0.6062  0.5444
## ar2         0.5434    0.1758   3.0916  0.0020
## ar3         0.0659    0.0820   0.8028  0.4221
## ma1         0.7459    0.3337   2.2355  0.0254
## sar1        0.9919    0.0062 159.0018  0.0000
## sma1       -0.8574    0.0474 -18.1033  0.0000
## intercept  -0.0001    0.0111  -0.0112  0.9911
##
## Significance Flags:
##           Significant
## ar1                No
## ar2               Yes
## ar3                No
## ma1               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000679
## Min MA Root Modulus:  1.012908
```



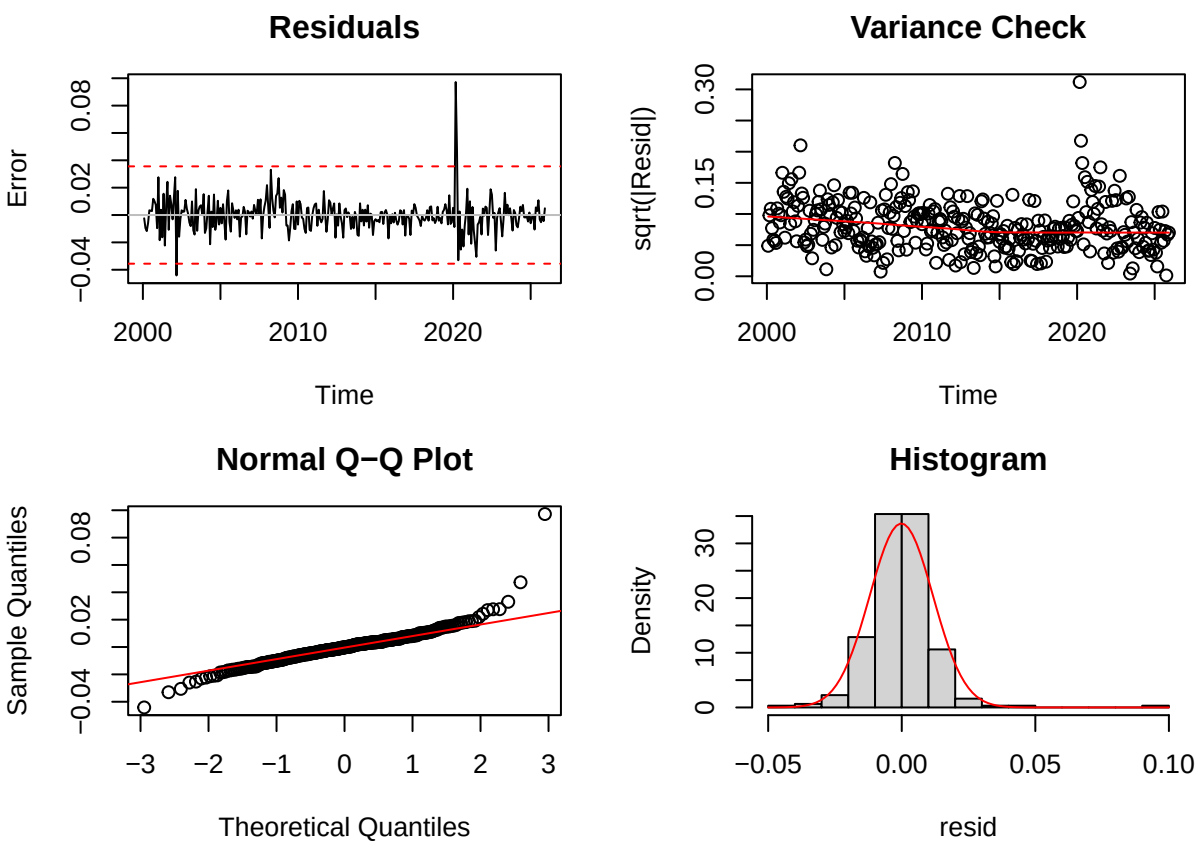

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(3,0,1)(1,0,1)12"
```

```
## AIC             "-1830.253"
## BIC             "-1800.334"
## Sigma2          "0.0001448423"
## Resid_Norm_PVal "9.04927e-15"
## Normal          "No"
## Resid_Ind_PVal  "0.3618883"
## Independent     "Yes"
## Min_AR_Root     "1.000679"
## Causal          "Yes"
## Min_MA_Root     "1.012908"
## Invertible      "Yes"
## RMSE            "0.005984126"
## RMSPE           "0.6048544"
## MAE             "0.004806914"
## MAPE_Pct        "0.4838689"
## Mean_CI_Width   "0.0589661"
##
## Processing Model 4 of 15...
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(4,0,1)(1,0,1)12
## ================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         1.1693    0.1690   6.9195  0.0000
## ar2        -0.2201    0.1204  -1.8290  0.0674
## ar3        -0.1669    0.0869  -1.9206  0.0548
## ar4         0.1340    0.0594   2.2556  0.0241
## ma1        -0.6475    0.1627  -3.9793  0.0001
## sar1        0.9935    0.0050 198.5790  0.0000
## sma1       -0.8679    0.0443 -19.5846  0.0000
## intercept  -0.0006    0.0164  -0.0396  0.9684
##
## Significance Flags:
##           Significant
## ar1               Yes
## ar2                No
## ar3                No
## ar4               Yes
## ma1               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000541
## Min MA Root Modulus:  1.011875
```

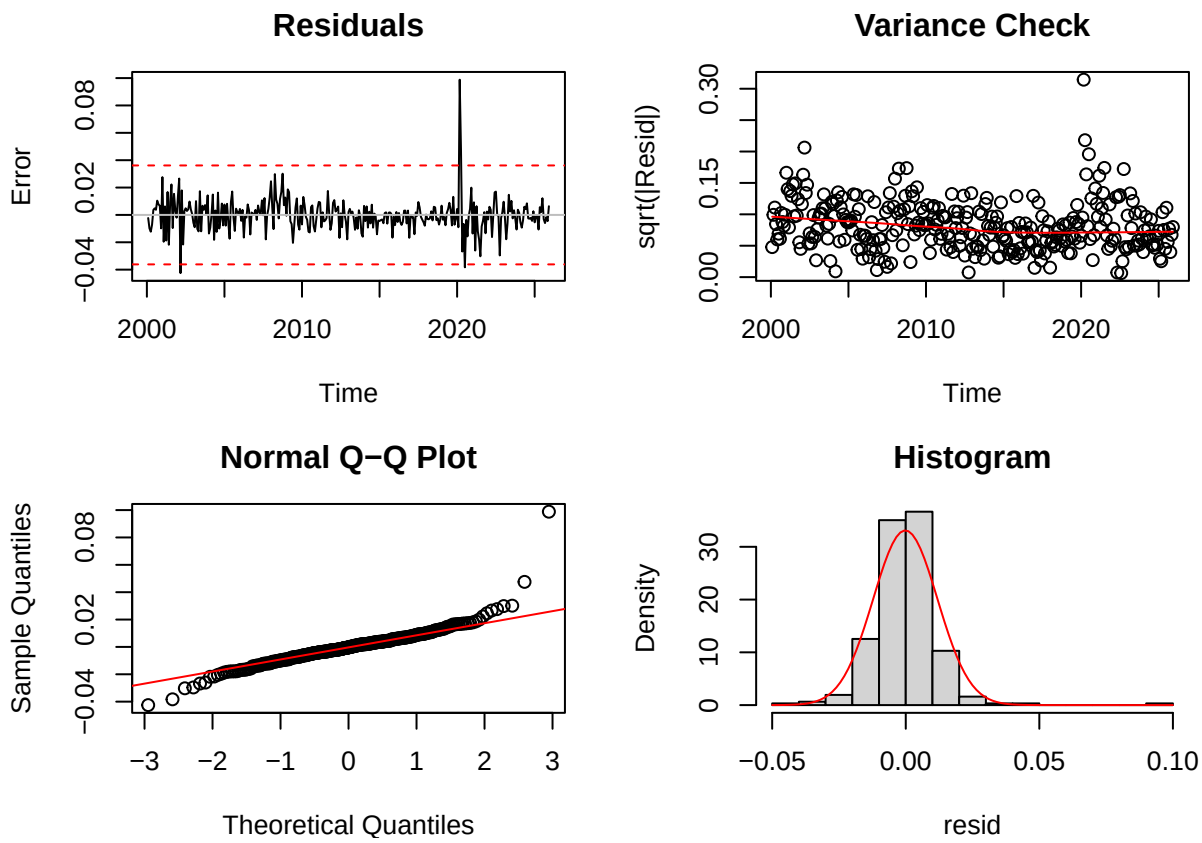



```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(4,0,1)(1,0,1)12"
## AIC              "-1836.007"
## BIC              "-1802.349"
## Sigma2           "0.0001404706"
## Resid_Norm_PVal "6.731279e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.6972193"
## Independent      "Yes"
## Min_AR_Root      "1.000541"
## Causal           "Yes"
## Min_MA_Root      "1.011875"
## Invertible       "Yes"
## RMSE             "0.006001468"
## RMSPE            "0.6056805"
## MAE              "0.004795174"
## MAPE_Pct         "0.4822412"
## Mean_CI_Width    "0.05701489"
##
## Processing Model 5 of 15...
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(5,0,1)(1,0,1)12
## ================================================================
##
```

```
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.7046    0.4588   1.5360  0.1245
## ar2         0.0335    0.2551   0.1312  0.8956
## ar3        -0.1096    0.0873  -1.2560  0.2091
## ar4         0.0510    0.0793   0.6429  0.5203
## ar5         0.1102    0.0824   1.3381  0.1809
## ma1        -0.1834    0.4623  -0.3967  0.6916
## sar1        0.9933    0.0052 192.8155  0.0000
## sma1       -0.8653    0.0447 -19.3764  0.0000
## intercept   0.0001    0.0152   0.0044  0.9965
##
## Significance Flags:
##           Significant
## ar1                No
## ar2                No
## ar3                No
## ar4                No
## ar5                No
## ma1                No
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000561
## Min MA Root Modulus:  1.012133
```

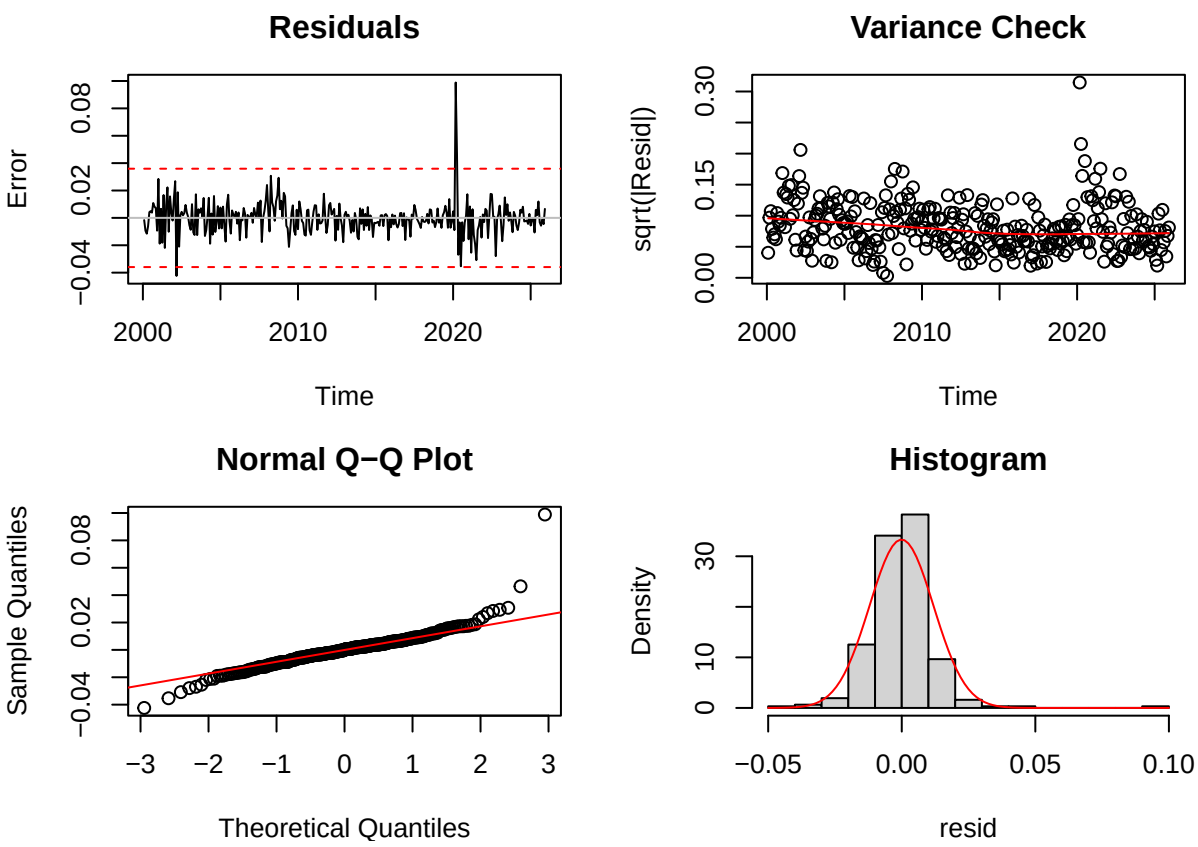


```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(5,0,1)(1,0,1)12"
## AIC              "-1834.788"
## BIC              "-1797.39"
## Sigma2           "0.0001402015"
## Resid_Norm_PVal "9.262329e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.7424442"
## Independent      "Yes"
## Min_AR_Root      "1.000561"
## Causal           "Yes"
## Min_MA_Root      "1.012133"
## Invertible       "Yes"
## RMSE             "0.005917737"
## RMSPE            "0.5970759"
## MAE              "0.004740685"
## MAPE_Pct         "0.4767073"
## Mean_CI_Width    "0.05733844"
##
## Processing Model 6 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(1,0,2)(1,0,1)12
## ==================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.8264    0.1030   8.0200  0.0000
## ma1        -0.3152    0.1400  -2.2515  0.0244
## ma2        -0.0811    0.1045  -0.7766  0.4374
## sar1        0.9919    0.0061 163.5403  0.0000
## sma1       -0.8537    0.0466 -18.3364  0.0000
## intercept  -0.0004    0.0135  -0.0325  0.9741
##
## Significance Flags:
##           Significant
## ar1               Yes
## ma1               Yes
## ma2                No
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000682
## Min MA Root Modulus:  1.013272
```



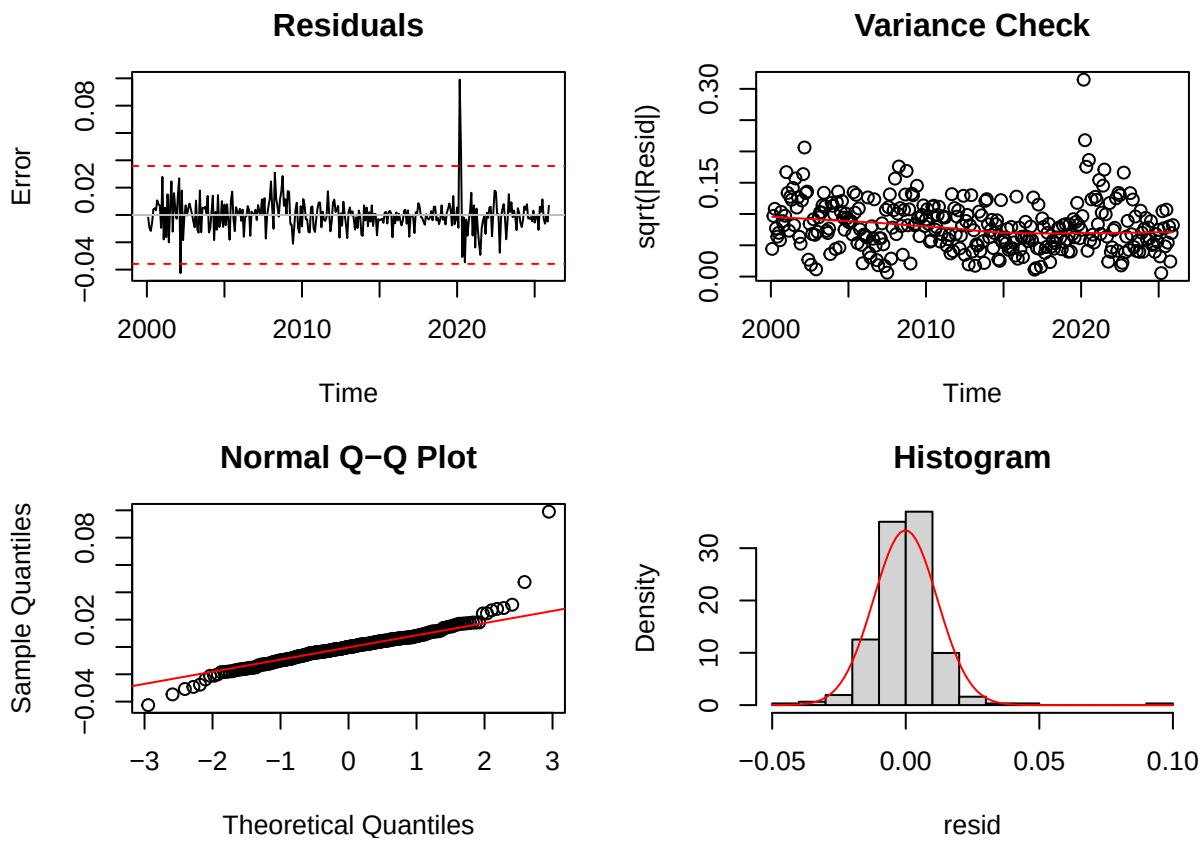

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(1,0,2)(1,0,1)12"
## AIC              "-1832.079"
## BIC              "-1805.901"
## Sigma2           "0.0001447888"
## Resid_Norm_PVal "9.770616e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.3321379"
## Independent      "Yes"
## Min_AR_Root      "1.000682"
## Causal           "Yes"
## Min_MA_Root      "1.013272"
## Invertible       "Yes"
## RMSE             "0.005861554"
## RMSPE            "0.5922128"
## MAE              "0.004647917"
## MAPE_Pct         "0.467709"
## Mean_CI_Width    "0.05804644"
##
## Processing Model 7 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(2,0,2)(1,0,1)12
## ==================================================================
##
```

```
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         1.3550    0.2015   6.7236  0.0000
## ar2        -0.4057    0.1729  -2.3456  0.0190
## ma1        -0.8339    0.2020  -4.1288  0.0000
## ma2         0.0415    0.1060   0.3914  0.6955
## sar1        0.9929    0.0055 179.6416  0.0000
## sma1       -0.8639    0.0458 -18.8728  0.0000
## intercept  -0.0015    0.0158  -0.0969  0.9228
##
## Significance Flags:
##           Significant
## ar1               Yes
## ar2               Yes
## ma1               Yes
## ma2                No
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000597
## Min MA Root Modulus:  1.012262
```



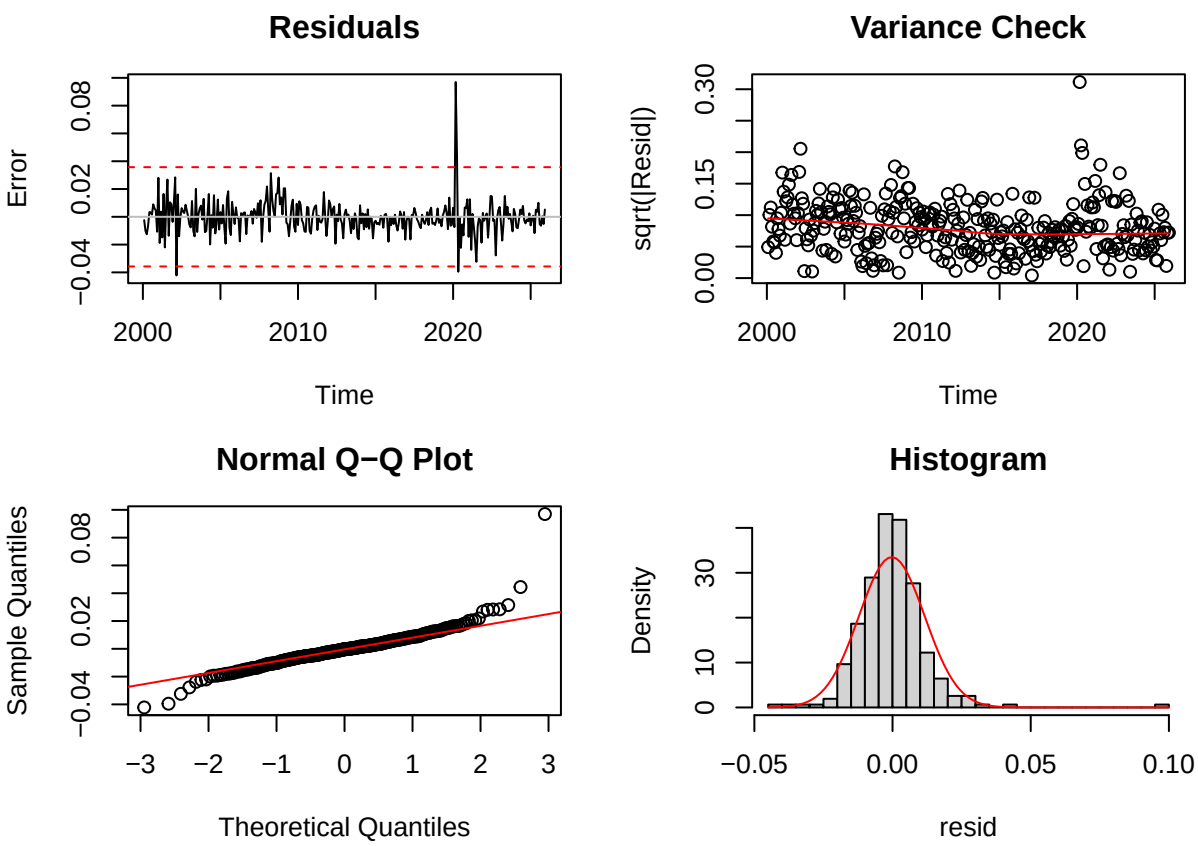

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(2,0,2)(1,0,1)12"
```

```
## AIC             "-1833.228"
## BIC             "-1803.31"
## Sigma2          "0.0001430103"
## Resid_Norm_PVal "8.844384e-15"
## Normal          "No"
## Resid_Ind_PVal  "0.4780029"
## Independent     "Yes"
## Min_AR_Root     "1.000597"
## Causal          "Yes"
## Min_MA_Root     "1.012262"
## Invertible      "Yes"
## RMSE            "0.005894541"
## RMSPE           "0.5951701"
## MAE             "0.004670329"
## MAPE_Pct        "0.4698026"
## Mean_CI_Width   "0.05741869"
##
## Processing Model 8 of 15...
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(3,0,2)(1,0,1)12
## ================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.5836    0.3011   1.9384  0.0526
## ar2         0.6497    0.3285   1.9780  0.0479
## ar3        -0.3206    0.1116  -2.8730  0.0041
## ma1        -0.0704    0.3021  -0.2330  0.8158
## ma2        -0.5708    0.2173  -2.6265  0.0086
## sar1        0.9931    0.0053 187.5410  0.0000
## sma1       -0.8648    0.0451 -19.1797  0.0000
## intercept  -0.0008    0.0160  -0.0484  0.9614
##
## Significance Flags:
##           Significant
## ar1                No
## ar2               Yes
## ar3               Yes
## ma1                No
## ma2               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000574
## Min MA Root Modulus:  1.012173
```



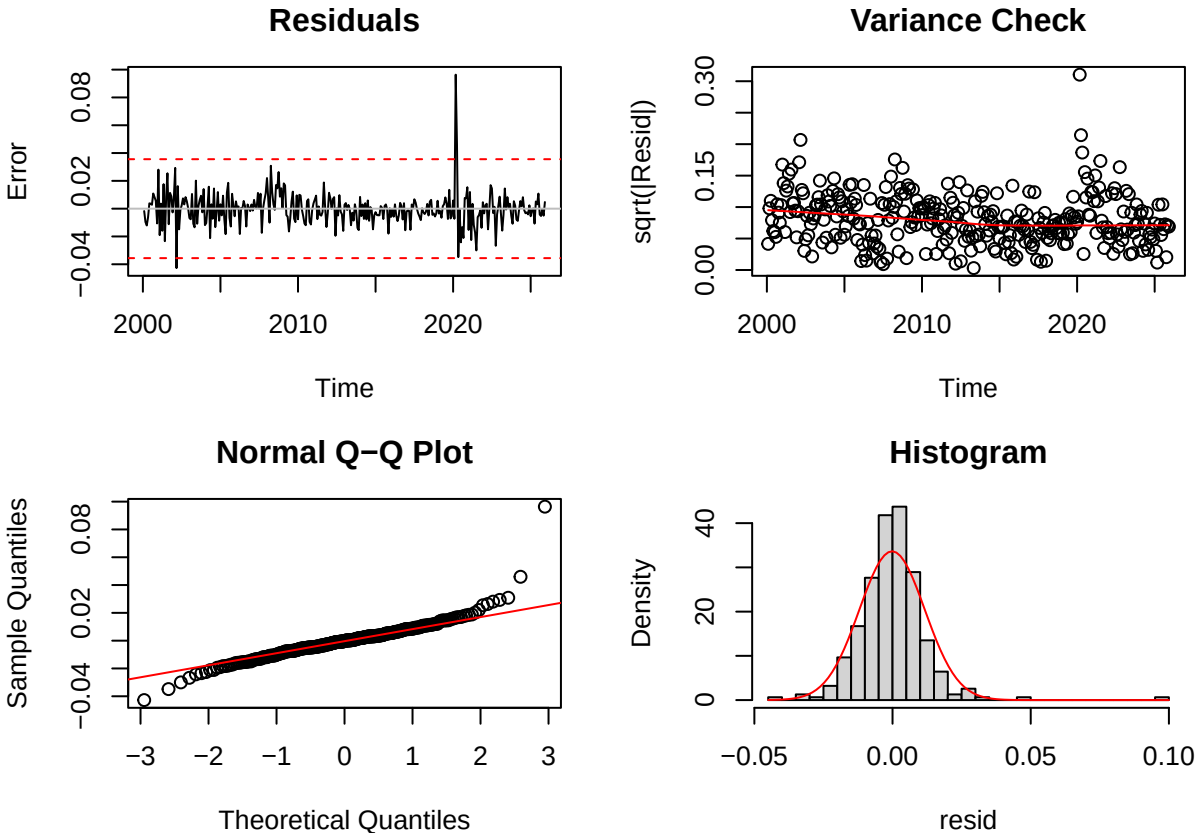

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(3,0,2)(1,0,1)12"
## AIC              "-1833.005"
## BIC              "-1799.347"
## Sigma2           "0.0001420242"
## Resid_Norm_PVal "4.884623e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.5779688"
## Independent      "Yes"
## Min_AR_Root      "1.000574"
## Causal           "Yes"
## Min_MA_Root      "1.012173"
## Invertible       "Yes"
## RMSE             "0.005922708"
## RMSPE            "0.5980141"
## MAE              "0.004684832"
## MAPE_Pct         "0.4712502"
## Mean_CI_Width    "0.05740074"
##
## Processing Model 9 of 15...
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(4,0,2)(1,0,1)12
## ================================================================
##
```

```
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.2166    0.1366   1.5851  0.1130
## ar2        -0.5172    0.1080  -4.7913  0.0000
## ar3         0.4244    0.1204   3.5242  0.0004
## ar4         0.1212    0.0637   1.9035  0.0570
## ma1         0.3334    0.1228   2.7154  0.0066
## ma2         0.8806    0.1154   7.6282  0.0000
## sar1        0.9926    0.0059 169.0281  0.0000
## sma1       -0.8672    0.0461 -18.8139  0.0000
## intercept  -0.0004    0.0108  -0.0362  0.9711
##
## Significance Flags:
##           Significant
## ar1                No
## ar2               Yes
## ar3               Yes
## ar4                No
## ma1               Yes
## ma2               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000619
## Min MA Root Modulus:  1.011946
```



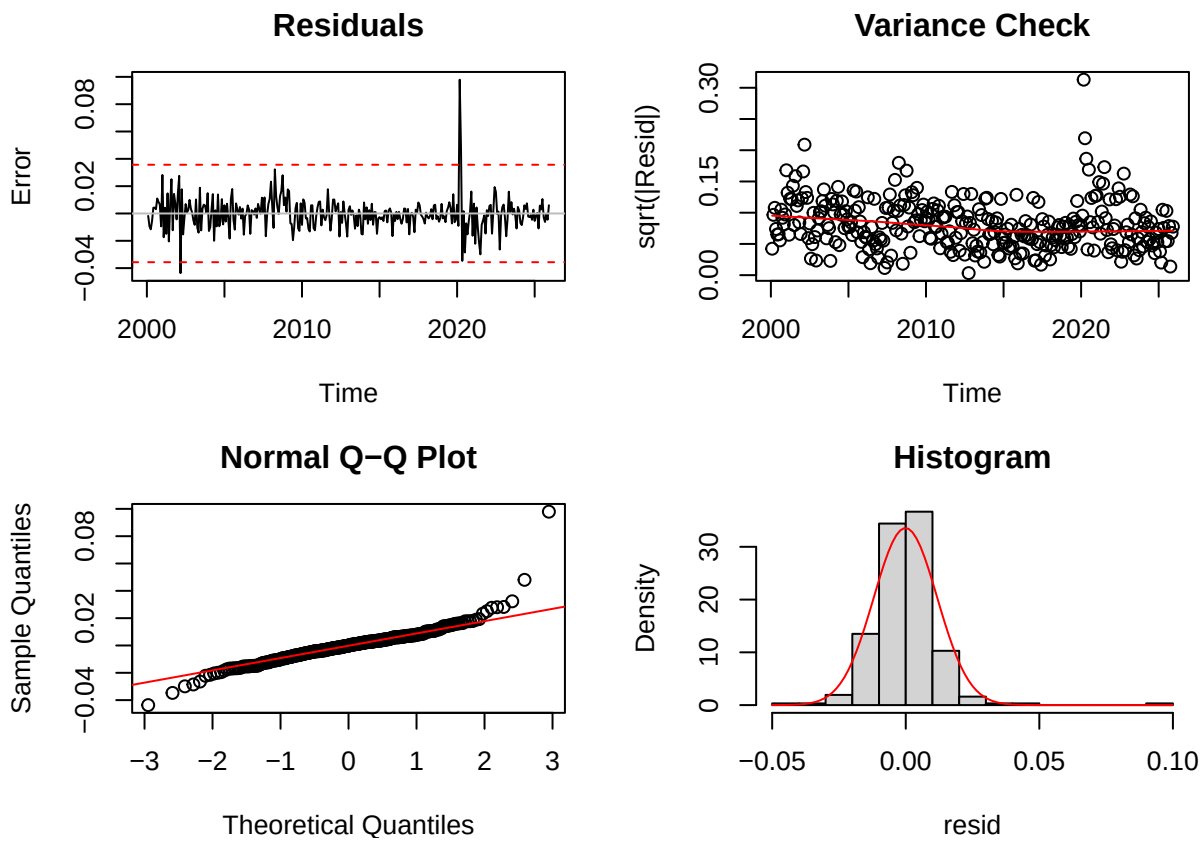
```
## Warning in arima(train_set, order = order, seasonal = seasonal): possible
## convergence problem: optim gave code = 1

##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(4,0,2)(1,0,1)12"
## AIC              "-1832.958"
## BIC              "-1795.56"
## Sigma2           "0.0001415366"
## Resid_Norm_PVal "1.284736e-14"
## Normal           "No"
## Resid_Ind_PVal  "0.6870367"
## Independent      "Yes"
## Min_AR_Root      "1.000619"
## Causal           "Yes"
## Min_MA_Root      "1.011946"
## Invertible       "Yes"
## RMSE             "0.006196291"
## RMSPE            "0.6263256"
## MAE              "0.004882362"
## MAPE_Pct         "0.4911521"
## Mean_CI_Width   "0.05867902"
##
## Processing Model 10 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(5,0,2)(1,0,1)12
## ==================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.2925    0.2038   1.4353  0.1512
## ar2        -0.3383    0.2630  -1.2866  0.1982
## ar3         0.2835    0.1981   1.4310  0.1524
## ar4         0.0796    0.0724   1.1004  0.2712
## ar5         0.1237    0.0686   1.8041  0.0712
## ma1         0.2441    0.2032   1.2016  0.2295
## ma2         0.6180    0.2996   2.0631  0.0391
## sar1        0.9923    0.0059 168.3074  0.0000
## sma1       -0.8602    0.0470 -18.3070  0.0000
## intercept  -0.0016    0.0126  -0.1239  0.9014
##
## Significance Flags:
##           Significant
## ar1                No
## ar2                No
## ar3                No
## ar4                No
## ar5                No
## ma1                No
## ma2               Yes
## sar1              Yes
## sma1              Yes
```

```
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000641
## Min MA Root Modulus:  1.012633
```



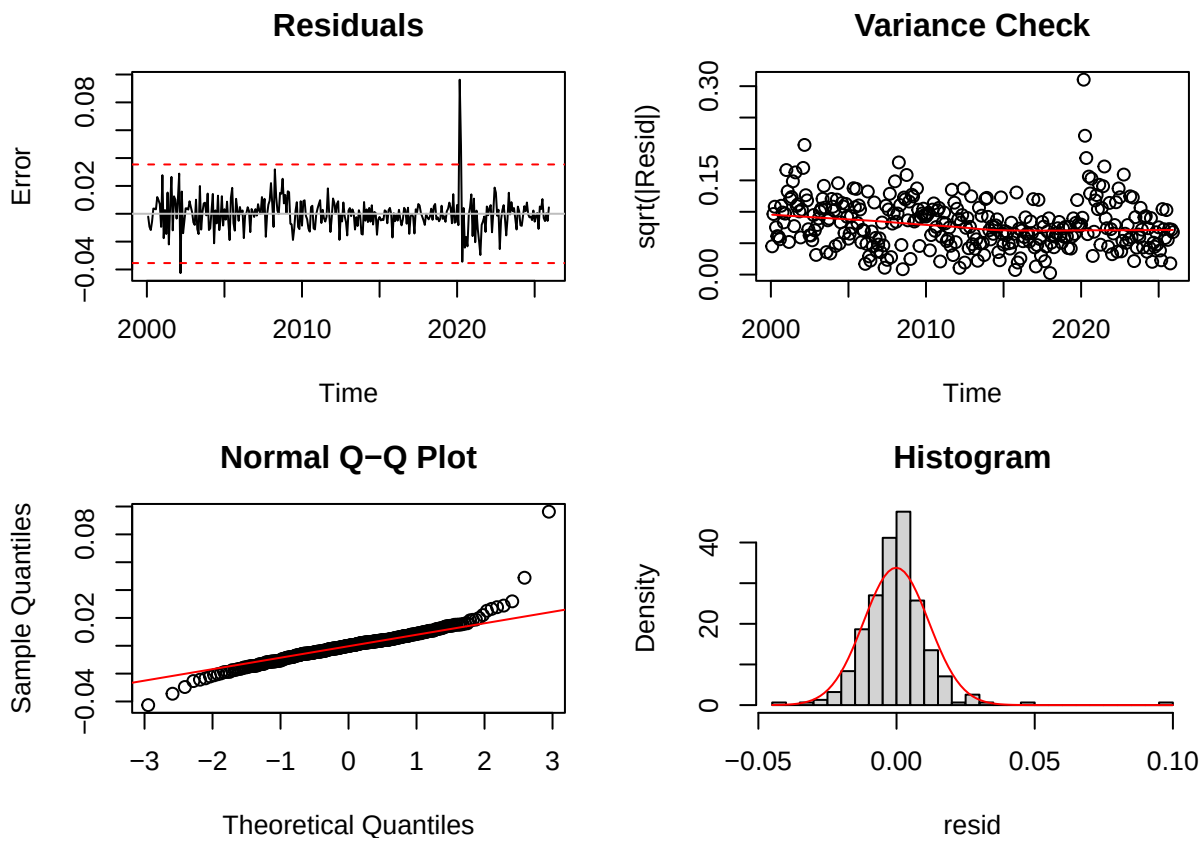

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(5,0,2)(1,0,1)12"
## AIC              "-1833.762"
## BIC              "-1792.624"
## Sigma2           "0.000140209"
## Resid_Norm_PVal "2.311044e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.8359843"
## Independent      "Yes"
## Min_AR_Root      "1.000641"
## Causal           "Yes"
## Min_MA_Root      "1.012633"
## Invertible       "Yes"
## RMSE             "0.005867103"
## RMSPE            "0.5923457"
## MAE              "0.004720548"
## MAPE_Pct         "0.4749116"
## Mean_CI_Width    "0.05830403"
##
## Processing Model 11 of 15...
```

```
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(1,0,3)(1,0,1)12
## ==================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.9123    0.0440  20.7500  0.0000
## ma1        -0.3955    0.0721  -5.4844  0.0000
## ma2        -0.0575    0.0701  -0.8206  0.4119
## ma3        -0.1783    0.0657  -2.7122  0.0067
## sar1        0.9935    0.0051 194.6336  0.0000
## sma1       -0.8696    0.0447 -19.4694  0.0000
## intercept  -0.0011    0.0162  -0.0668  0.9467
##
## Significance Flags:
##           Significant
## ar1               Yes
## ma1               Yes
## ma2                No
## ma3               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000541
## Min MA Root Modulus:  1.011715
```



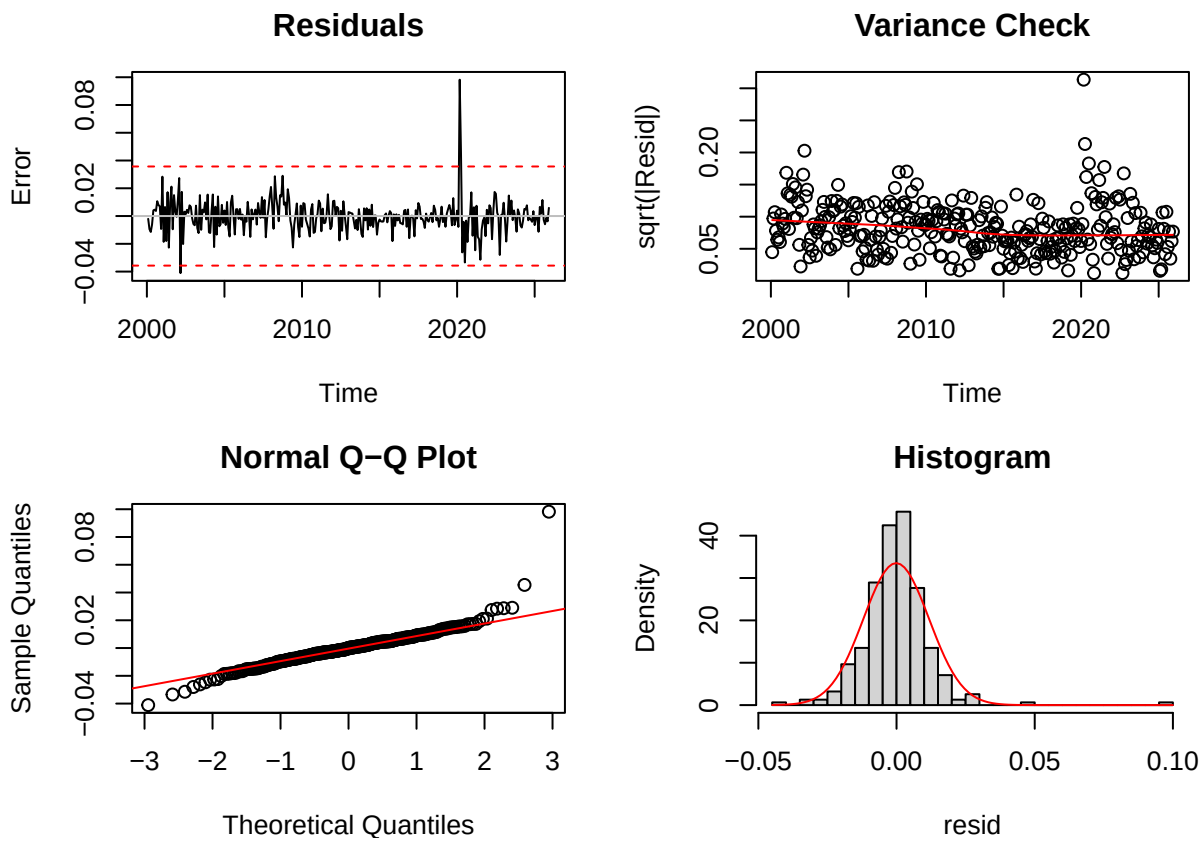

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(1,0,3)(1,0,1)12"
## AIC              "-1837.381"
## BIC              "-1807.463"
## Sigma2           "0.0001408264"
## Resid_Norm_PVal "5.37119e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.683126"
## Independent      "Yes"
## Min_AR_Root      "1.000541"
## Causal           "Yes"
## Min_MA_Root      "1.011715"
## Invertible       "Yes"
## RMSE             "0.005929427"
## RMSPE            "0.5985817"
## MAE              "0.004714886"
## MAPE_Pct         "0.4742491"
## Mean_CI_Width    "0.05732075"
##
## Processing Model 12 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(2,0,3)(1,0,1)12
## ==================================================================
##   Error in Model 12: non-finite finite-difference value [2]
```

```
##
## Processing Model 13 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(3,0,3)(1,0,1)12
## ==================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.9588    0.2356   4.0699  0.0000
## ar2        -0.5675    0.2111  -2.6877  0.0072
## ar3         0.4492    0.1659   2.7084  0.0068
## ma1        -0.4508    0.2247  -2.0064  0.0448
## ma2         0.5015    0.1917   2.6155  0.0089
## ma3        -0.4022    0.0912  -4.4126  0.0000
## sar1        0.9932    0.0052 190.5073  0.0000
## sma1       -0.8644    0.0451 -19.1835  0.0000
## intercept  -0.0006    0.0158  -0.0386  0.9692
##
## Significance Flags:
##           Significant
## ar1               Yes
## ar2               Yes
## ar3               Yes
## ma1               Yes
## ma2               Yes
## ma3               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000567
## Min MA Root Modulus:  1.012216
```



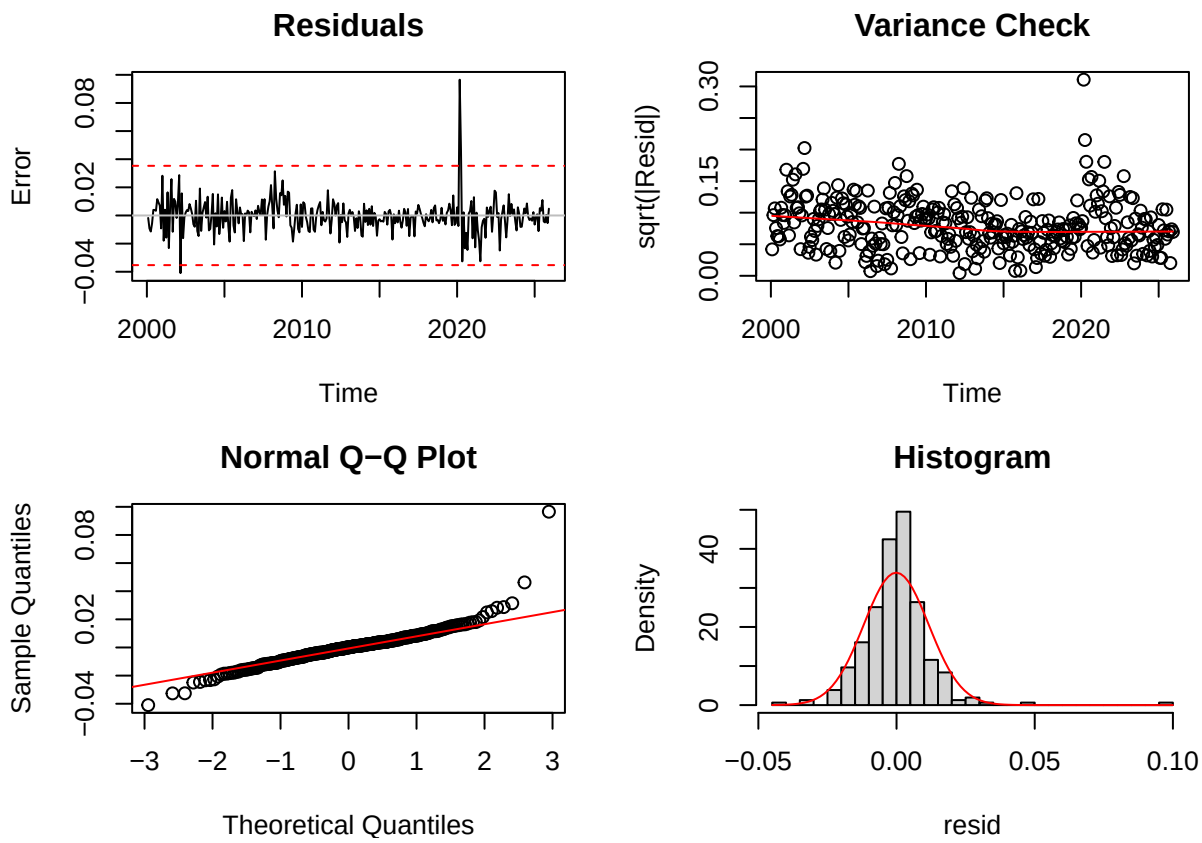

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(3,0,3)(1,0,1)12"
## AIC              "-1837.647"
## BIC              "-1800.249"
## Sigma2           "0.0001388844"
## Resid_Norm_PVal "1.043977e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.8788881"
## Independent      "Yes"
## Min_AR_Root      "1.000567"
## Causal           "Yes"
## Min_MA_Root      "1.012216"
## Invertible       "Yes"
## RMSE             "0.005873528"
## RMSPE            "0.5927018"
## MAE              "0.004686809"
## MAPE_Pct         "0.4713145"
## Mean_CI_Width    "0.05711622"
##
## Processing Model 14 of 15...
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(4,0,3)(1,0,1)12
## ================================================================
##
```

```
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.0256    0.1276   0.2009  0.8408
## ar2         0.3459    0.0691   5.0062  0.0000
## ar3         0.8261    0.0319  25.8772  0.0000
## ar4        -0.3727    0.0979  -3.8052  0.0001
## ma1         0.5031    0.1084   4.6398  0.0000
## ma2        -0.0138    0.1311  -0.1054  0.9161
## ma3        -0.7650    0.1027  -7.4499  0.0000
## sar1        0.9929    0.0055 181.8875  0.0000
## sma1       -0.8616    0.0463 -18.6007  0.0000
## intercept  -0.0007    0.0161  -0.0409  0.9674
##
## Significance Flags:
##           Significant
## ar1                No
## ar2               Yes
## ar3               Yes
## ar4               Yes
## ma1               Yes
## ma2                No
## ma3               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000594
## Min MA Root Modulus:  1.008526
```




```
## Warning in arima(train_set, order = order, seasonal = seasonal): possible
## convergence problem: optim gave code = 1

##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(4,0,3)(1,0,1)12"
## AIC              "-1830.114"
## BIC              "-1788.977"
## Sigma2           "0.0001412112"
## Resid_Norm_PVal "1.581858e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.573673"
## Independent      "Yes"
## Min_AR_Root      "1.000594"
## Causal           "Yes"
## Min_MA_Root      "1.008526"
## Invertible       "Yes"
## RMSE             "0.006029664"
## RMSPE            "0.6088515"
## MAE              "0.00472935"
## MAPE_Pct         "0.4755219"
## Mean_CI_Width    "0.05722668"
##
## Processing Model 15 of 15...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(5,0,3)(1,0,1)12
```

```
## ====================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##           Estimate Std.Error   t_stat p_value
## ar1         0.9038    0.1896   4.7670  0.0000
## ar2        -0.6133    0.2310  -2.6552  0.0079
## ar3         0.7147    0.2810   2.5431  0.0110
## ar4        -0.1750    0.1286  -1.3600  0.1738
## ar5         0.0284    0.0808   0.3507  0.7258
## ma1        -0.3788    0.1806  -2.0980  0.0359
## ma2         0.5519    0.1722   3.2057  0.0013
## ma3        -0.5761    0.1982  -2.9058  0.0037
## sar1        0.9937    0.0050 197.0121  0.0000
## sma1       -0.8733    0.0449 -19.4651  0.0000
## intercept  -0.0012    0.0159  -0.0739  0.9411
##
## Significance Flags:
##           Significant
## ar1               Yes
## ar2               Yes
## ar3               Yes
## ar4                No
## ar5                No
## ma1               Yes
## ma2               Yes
## ma3               Yes
## sar1              Yes
## sma1              Yes
## intercept          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000526
## Min MA Root Modulus:  1.01135
```




```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(5,0,3)(1,0,1)12"
## AIC              "-1835.399"
## BIC              "-1790.521"
## Sigma2           "0.0001380309"
## Resid_Norm_PVal  "1.257154e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.9222191"
## Independent      "Yes"
## Min_AR_Root      "1.000526"
## Causal           "Yes"
## Min_MA_Root      "1.01135"
## Invertible       "Yes"
## RMSE             "0.006045785"
## RMSPE            "0.6096975"
## MAE              "0.004806207"
## MAPE_Pct         "0.4832333"
## Mean_CI_Width    "0.05703823"
##
## --- Comparison Complete ---
```

```r
print(comparison_table)
```

```
##                Model_Name       AIC       BIC       Sigma2 Resid_Norm_PVal
## 1  ARIMA(1,0,1)(1,0,1)12 -1833.625 -1811.186 0.0001452986    1.170265e-14
## 2  ARIMA(2,0,1)(1,0,1)12 -1835.071 -1808.892 0.0001429969    9.511915e-15
```

```
## 11 ARIMA(1,0,3)(1,0,1)12 -1837.381 -1807.463 0.0001408264    5.371190e-15
## 6  ARIMA(1,0,2)(1,0,1)12 -1832.079 -1805.901 0.0001447888    9.770616e-15
## 7  ARIMA(2,0,2)(1,0,1)12 -1833.228 -1803.310 0.0001430103    8.844384e-15
## 4  ARIMA(4,0,1)(1,0,1)12 -1836.007 -1802.349 0.0001404706    6.731279e-15
## 3  ARIMA(3,0,1)(1,0,1)12 -1830.253 -1800.334 0.0001448423    9.049270e-15
## 12 ARIMA(3,0,3)(1,0,1)12 -1837.647 -1800.249 0.0001388844    1.043977e-14
## 8  ARIMA(3,0,2)(1,0,1)12 -1833.005 -1799.347 0.0001420242    4.884623e-15
## 5  ARIMA(5,0,1)(1,0,1)12 -1834.788 -1797.390 0.0001402015    9.262329e-15
## 9  ARIMA(4,0,2)(1,0,1)12 -1832.958 -1795.560 0.0001415366    1.284736e-14
## 10 ARIMA(5,0,2)(1,0,1)12 -1833.762 -1792.624 0.0001402090    2.311044e-14
## 14 ARIMA(5,0,3)(1,0,1)12 -1835.399 -1790.521 0.0001380309    1.257154e-14
## 13 ARIMA(4,0,3)(1,0,1)12 -1830.114 -1788.977 0.0001412112    1.581858e-14
##    Normal Resid_Ind_PVal Independent Min_AR_Root Causal Min_MA_Root Invertible
## 1      No      0.3330666         Yes    1.000721    Yes    1.013275        Yes
## 2      No      0.4642005         Yes    1.000582    Yes    1.012163        Yes
## 11     No      0.6831260         Yes    1.000541    Yes    1.011715        Yes
## 6      No      0.3321379         Yes    1.000682    Yes    1.013272        Yes
## 7      No      0.4780029         Yes    1.000597    Yes    1.012262        Yes
## 4      No      0.6972193         Yes    1.000541    Yes    1.011875        Yes
## 3      No      0.3618883         Yes    1.000679    Yes    1.012908        Yes
## 12     No      0.8788881         Yes    1.000567    Yes    1.012216        Yes
## 8      No      0.5779688         Yes    1.000574    Yes    1.012173        Yes
## 5      No      0.7424442         Yes    1.000561    Yes    1.012133        Yes
## 9      No      0.6870367         Yes    1.000619    Yes    1.011946        Yes
## 10     No      0.8359843         Yes    1.000641    Yes    1.012633        Yes
## 14     No      0.9222191         Yes    1.000526    Yes    1.011350        Yes
## 13     No      0.5736730         Yes    1.000594    Yes    1.008526        Yes
##           RMSE     RMSPE         MAE  MAPE_Pct Mean_CI_Width
## 1  0.005897894 0.5961513 0.004715277 0.4746397    0.05873444
## 2  0.005911938 0.5968776 0.004688363 0.4715944    0.05737218
## 11 0.005929427 0.5985817 0.004714886 0.4742491    0.05732075
## 6  0.005861554 0.5922128 0.004647917 0.4677090    0.05804644
## 7  0.005894541 0.5951701 0.004670329 0.4698026    0.05741869
## 4  0.006001468 0.6056805 0.004795174 0.4822412    0.05701489
## 3  0.005984126 0.6048544 0.004806914 0.4838689    0.05896610
## 12 0.005873528 0.5927018 0.004686809 0.4713145    0.05711622
## 8  0.005922708 0.5980141 0.004684832 0.4712502    0.05740074
## 5  0.005917737 0.5970759 0.004740685 0.4767073    0.05733844
## 9  0.006196291 0.6263256 0.004882362 0.4911521    0.05867902
## 10 0.005867103 0.5923457 0.004720548 0.4749116    0.05830403
## 14 0.006045785 0.6096975 0.004806207 0.4832333    0.05703823
## 13 0.006029664 0.6088515 0.004729350 0.4755219    0.05722668
```

4 Choosing Model

All modeels dont have normal residuals because of covid. Disregarding that, based on model performance we have models 13, 3, 6 but for parsimony were choosing the simplest which is ARIMA(1,0,2)(1,0,1)[12].